Gradient descent on Boolean targets as noise-limited greedy feature growth: experiments, discrete models, and generalization bounds

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Abstract

We look for a low-level, discrete description of what gradient descent does when it trains a small network in the feature-learning regime, analogous to the correspondence between Bayesian neural networks and Occam circuit learning. The testbed is a μP two-hidden-layer ReLU network trained by Adam on Boolean targets, with input dimension $d \in \{15, 30, 60, 120\}$. We report three things.

Phenomenology. The training-set size n^* at which test error reaches 0.1 grows polynomially in d, with an exponent set by how many inputs must be found at once. For k-bit parities it grows like $d^{1.0}$, $d^{1.9}$ and $d^{2.4}$ for $k = 2, 3, 4$. This is close to the correlational-SQ rate d^{k-1} for $k \leq 3$, and far from the $k \log d$ an Occam learner needs. The ratio of the networks' n^* to an Occam learner's grows with the target's leap, from about $8\times$ at leap 1 to $85\times$ at leap 4.

A discrete model. A model that grows a circuit one feature at a time, finding each candidate by a continuous search on the training sample (the race model), reproduces the networks' n^* to within about $1.15\times$ on 20 random formulas it was not calibrated on. Ablations show that the sampling noise of this search, not the model's race dynamics, sets n^*. They suggest a simpler rule: accept a candidate feature once its correlation with the residual beats that noise. A two-constant closed form of this rule predicted the networks' 3-parity sample complexity at $d = 120$ to within $1.2\times$ before those networks were trained; the race model was $3\times$ too optimistic there. On partially learned tasks, the networks' individual mistakes look like those of their initial kernel. About 45% of the output variance is kernel-shaped fitted sample noise, which suggests the picture "random-feature kernel background plus a few grown features".

Bounds. We prove versions of "short training ⇒ small circuit ⇒ generalization" for:

- the discrete model, with size read as depth, incoherence-weighted size, or readout norm at depth one;
- diagonal linear networks;
- two-layer quadratic networks with fixed output signs, where the number of grown units is at most $\kappa(\tau/\Theta)^2$ uniformly in time;
- two-layer ReLU networks under gradient flow, through the first escape from the initial saddle.

*Written by an AI system in a series of sessions prompted and steered by Kaarel Hänni. Not reviewed by a human. The proofs were checked by independent adversarial AI referees, and the numbers by independent AI fact-checkers, as described in Appendix D. Code, generated reports and summary data, derivations and referee reports are on the `claude/nifty-mccarthy-j0lw2a` branch of https://github.com/kaarelh/AI-works, folder `gd-circuit-growth/`.

On the real networks, a-priori bounds are vacuous or weak. Distillation certificates ("the network agrees on unlabelled inputs with a small circuit") give non-vacuous bounds on 221 of 315 trained networks, and are never violated.

Finally, we test a question about compositional structure that leap complexity cannot see: does $s + x_1x_2 + x_3x_4 + x_1x_2x_3x_4$ (with $s = x_1 + \cdots + x_4$) learn its degree-4 term faster than $s + x_1x_2x_3x_4$? We recorded predictions before inspecting any network result (two controls were added afterwards). It does: by 1.2× in samples at $d = 30$ and 1.6× at $d = 60$, and by up to 10× in training time (at $n = 2048$). Any intermediate feature on the term's own inputs helps, while pairs on other inputs do not. This suggests that the cost of a move is set by how many existing features it must combine, not by how many new inputs it needs.

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1 Introduction

For Bayesian neural networks there is a useful low-level picture of generalization, close to Occam learning over small circuits [11]. In the strong-prior limit, and at mean-field or stronger scaling ($\gamma \geq 1$), the posterior learns any target computed by a small circuit, with sample cost polynomial in the input dimension and the circuit's size. Below that scaling ($\gamma < 1$) it can need exponentially many samples. This paper asks whether gradient descent admits a comparably concrete, discrete description, and how close it is to the Occam picture.

The literature already contains the qualitative ingredients. In the rich regime, from small initialization, gradient flow passes near a sequence of saddles. Each escape adds roughly one unit or one rank, so training resembles a greedy algorithm that grows a network one feature at a time. This has been made precise, to different degrees, in several settings:

- proved for diagonal linear networks [24], and under assumptions for matrix factorization [22];
- proved for the first escape in deep linear networks [18] and deep ReLU networks [5], with the later saddles conjectured;
- proposed as a general description of two-layer networks, "alternating gradient flows" [20].

For sparse Boolean targets, the *leap complexity* of Abbe, Boix-Adserà and Misiakiewicz [2, 3] predicts that online SGD on a two-layer network needs on the order of $d^{\max(L-1,1)}$ steps. Here L is the largest number of new coordinates any Fourier monomial adds when the monomials are taken in the best order. This matches correlational-SQ (CSQ) lower bounds [3]. For Gaussian single-index models with information exponent k, plain online SGD has the analogous rate d^{k-1} [6]. Evolution by mutation and selection is equivalent to CSQ learning [17, 25], and noisy neuroevolution is equivalent on average to gradient descent [26]. So the original idea behind this work, "a single circuit, mutated, keeping improvements", is already placed: it is CSQ learning.

What we add is quantitative and empirical. We train rich-regime networks on 41 Boolean targets (21 designed, 20 random formulas) at up to four input dimensions. We then build discrete models that reproduce the networks' sample complexity target by target, out of sample, and use ablations to find which ingredient of the model matters. We also prove a set of "training has not taken long, so the circuit is still small, so a union bound over small circuits applies" theorems, and test what can actually be certified for the trained networks.

Main findings.

1. **Gradient descent pays a search-noise cost, not a description-length cost** (Section 3). The networks' n^* grows like $d^{1.04}$, $d^{1.91}$ and $d^{2.40}$ for 2-, 3- and 4-bit parities (Table 2). An Occam learner needs about $k \log d$ samples.
2. **A greedy, noise-limited growth model matches the networks** (Section 4).
   - *The rule.* A candidate feature is accepted once $n\rho^2$ exceeds the noise level of the continuous search that would find it. Here ρ is its correlation with the residual, and the noise level grows by about a factor d/a for each new input beyond the second.
   - *Simulation.* The race model, a simulation of feature growth from which the rule was distilled, matches n^* on 20 fresh random formulas within 1.15× on average. The rule itself, simulated as a minimal model, is within about 1.4×.
   - *Closed form.* Evaluated in closed form along the target's greedy path, the rule is within about 1.34× on the same formulas, and it predicted the 3-parity at $d = 120$ before the networks were run.

- *Ablations.* The race dynamics we originally built into the model matter little: on held-out targets, removing them moves n^* by at most about 0.1 beyond the model's own spread, with one exception. The finite-sample noise of the direction search does matter.

3. **Memorize like a kernel, generalize like a feature learner** (Section 4.4; a [Heuristic] reading of [Numerical] results). On partially learned cells, the networks' mistakes are best predicted, among the models we tested, by the NNGP kernel of their architecture. A model that adds a kernel background to the grown features needs about 2 grown gates per such cell, where the pure growth model needs about 22.

4. **Bounds** (Section 6, Appendices A and B). The "short training ⇒ small" implication is a theorem for:

   - the discrete model, for size read as depth, incoherence-weighted size, or readout norm (at depth one), though not for the raw gate count;
   - diagonal linear networks;
   - quadratic networks with fixed output signs;
   - two-layer ReLU networks through the first escape.

   It is a conjecture beyond the first escape. On the real networks, the only tight bounds are a-posteriori distillation certificates.

5. **Composition** (Section 5). Leap complexity depends only on Fourier supports. In a test whose predictions were recorded before any network result was inspected, intermediate features on a degree-4 term's own inputs make that term cheaper. They do so by 1.2–1.6× in samples, and by up to 10× in time, between targets of the same leap, and the gap grows with d. Features on other inputs do not help.

Scope. Everything empirical here is one architecture family, a μP two-hidden-layer ReLU MLP, mostly at width 128 and with Adam. Inputs are uniform on the hypercube, every target has leap at most 4 (parities of at most 4 bits), and the loss is square loss. We label every claim [Proved], [Conditional], [Conjecture], [Heuristic] or [Numerical].

2 Setup

Networks. The network is $f(x) = a^\top \mathrm{relu}(W_2\,\mathrm{relu}(W_1 x + b_1) + b_2)$ with two hidden layers of width $N = 128$ (32–256 in Section 3.4). It uses the μP (maximal-update) parameterization [27]:

- *Initialization:* $W_1 \sim \mathcal{N}(0, 1/d)$, $W_2 \sim \mathcal{N}(0, 1/N)$, a small readout $a \sim \mathcal{N}(0, 1/N^2)$, and zero biases.
- *Learning rates:* per-layer multipliers relative to a base width of 64.

Training minimizes the square loss on ± 1 labels with Adam [19], learning rate 10^{-3}. Minibatches of 64 are drawn with replacement from a fixed training set of size $n \in \{16, 32, \ldots, 16384\}$ (up to 65536 at $d = 60$) for 20000 steps. At $d = 120$ the n grids are target-specific, up to 65536, and the runs last 40000 steps, with n^* read at step 20000. There are 3 seeds for the designed targets at $d = 30$ (Adam, width 128), and 2 for the other dimensions, the random formulas, SGD and the other widths. At $n = 16384$ the weights move far from initialization, $\|\Delta W_1\|/\|W_1(0)\| \approx 0.7$–$1.05$, and the participation-ratio ranks of the layer-1 and layer-2 kernels collapse from about 36 and 32 to 3–12 and 1–10. This is the feature-learning regime.

Two other protocols are used:

- *SGD:* μP-SGD with learning rate 0.2, on a subset of targets.
- *Online:* a fresh minibatch at every step, i.e. one pass over the data.

Targets. Inputs are uniform on $\{\pm 1\}^d$. There are 21 designed targets:

- a dictator x_1; AND_3; MAJ_3 and MAJ_9; a random halfspace on 10 bits;
- the parities $\chi_k = x_1 \cdots x_k$ for $k = 2, 3, 4$;
- a 5-term staircase $\mathrm{sign}(x_1 + x_1x_2 + \dots)$;
- compositions: MAJ of parities, parity of majorities, recursive majority, $\mathrm{AND}(x_1x_2, x_3x_4)$;
- a multiplexer, a 3-term DNF, a decision list, $\mathrm{sign}(2x_1 + x_2x_3 + x_4x_5x_6)$;
- four random 6-gate formulas.

They are split into 12 *calibration* targets and 9 *held-out* targets.[1] A further 20 random fan-in-2 AND/OR/XOR formulas on 4–5 inputs (3–5 of them relevant), with 3–8 gates, were generated after all model design and calibration, as an out-of-sample test. Section 5 uses real-valued polynomial targets.

Metrics. We use the following:

- **Sample complexity n^*:** the smallest n at which the seed-mean test error at the end of training is at most 0.1, interpolated linearly in $\log n$ on the factor-2 grid.
- **Comparing a model with the networks:**
  - the mean $|\log_2(n^*_{\text{model}}/n^*_{\text{NN}})|$ over targets; 0.33 means within a factor of $2^{0.33} \approx 1.26$ on average;
  - the Spearman correlation of n^*;
  - *success agreement*, the fraction of (f, n) cells on which model and networks agree about whether test error is at most 0.1;
  - the mean absolute difference of the test-error curves.
- **Per-example agreement:** computed on *partially learned cells*, where the networks' test error is in $[0.05, 0.45]$, with model and network trained on the same training set. It is the excess of their agreement on test points over the agreement that independent errors would give, as a fraction of the largest possible excess: 0 means independent errors, 1 means identical errors.
- **Time:** the Spearman correlation between a model's internal time to generalize and the networks' training steps to generalize.

Leap is the leap complexity of [3].

[1] Calibration: x_1, MAJ_3, the halfspace, χ_2, χ_4, the staircase, parity of majorities, the DNF, recursive majority, $\mathrm{AND}(x_1x_2, x_3x_4)$, and two random formulas. Held-out: AND_3, MAJ_9, χ_3, MAJ of parities, the multiplexer, the decision list, $\mathrm{sign}(2x_1 + x_2x_3 + x_4x_5x_6)$, and two random formulas.

target	relevant	leap	n^* networks	n^* NNGP kernel	n^* Occam (juntas)
x_1	1	1	47	29	16
MAJ_3	3	1	112	73	18
MAJ_9	9	1	263	1470	–
random halfspace, 10 bits	10	1	563	1701	–
χ_2, χ_3, χ_4	2, 3, 4	2, 3, 4	200, 843, 2728	406, –, –	$\leq$16, 23, 37
staircase, 5 terms	5	1	1514	–	–
$\mathrm{MAJ}(x_1x_2, x_3x_4, x_5x_6)$	6	2	1380	1129	–
multiplexer $x_4\,?\,x_1x_2x_3 : \mathrm{MAJ}_3$	4	1	1156	–	37
3-term DNF	9	1	505	564	–
$\mathrm{sign}(2x_1 + x_2x_3 + x_4x_5x_6)$	6	3	2205	–	–

Table 1: Sample complexity at $d = 30$, width 128, Adam (selected targets). "–" means test error never reached 0.1 in the range tested ($n \leq 4096$ for the kernel, $n \leq 16384$ otherwise). The junta learner cannot represent targets with more than 4 relevant variables.

3 What the networks do

3.1 Sample complexity at $d = 30$

Table 1 and Figure 1 show the networks' learning curves against two reference learners:

- the NNGP kernel [21] of the same architecture, in kernel ridge regression;
- an Occam learner that picks the smallest junta, of up to 4 variables, consistent with the training set.

The networks sit between the two, in a specific way [Numerical]:

- **Sparse high-order targets.** Parities cost the networks 3.2–4.2× more data per extra bit: 200, 843 and 2728 samples. The Occam learner needs at most 16, 23 and 37, and the kernel fails from 3 bits on.
- **Dense leap-1 targets.** MAJ_9 and the 10-bit halfspace are 3–6× cheaper for the networks than for the kernel, and out of reach of the junta learner.

Several further observations [Numerical]:

- **Reusing the training set saves data.** With fresh samples at every step, the networks need 12–64× their fixed-set n^*: about 3200 samples for MAJ_9, 12800 for χ_2, and 32000 for χ_3, the staircase, the multiplexer and MAJ of parities. These counts are read at the first checkpoint with test error at most 0.1. Checkpoints are 2–2.5× apart, so they are upper bounds that may be up to about 2.5× high.
- **Generalization is delayed near n^*.** At $n = 256$, χ_2 fits its training set by step 200 but generalizes only at steps 7000–10000. The multiplexer, the DNF, recursive majority and $\mathrm{AND}(x_1x_2, x_3x_4)$ show 4–40× delays at their critical n.
- **Features are learned in staircase order.** On the staircase target, the network's Fourier coefficient reaches half its target value at step 100 for x_1, step 500 for x_1x_2 and $x_1x_2x_3$, and step 1000 for $x_1x_2x_3x_4$.
- **Representation size peaks just below n^*.** For the 3- and 4-bit parities, the staircase and the multiplexer, the participation-ratio rank of the second layer's kernel peaks just below n^* and collapses above it. Below n^* the networks fit the training set with many features; above it they use the target's few features.

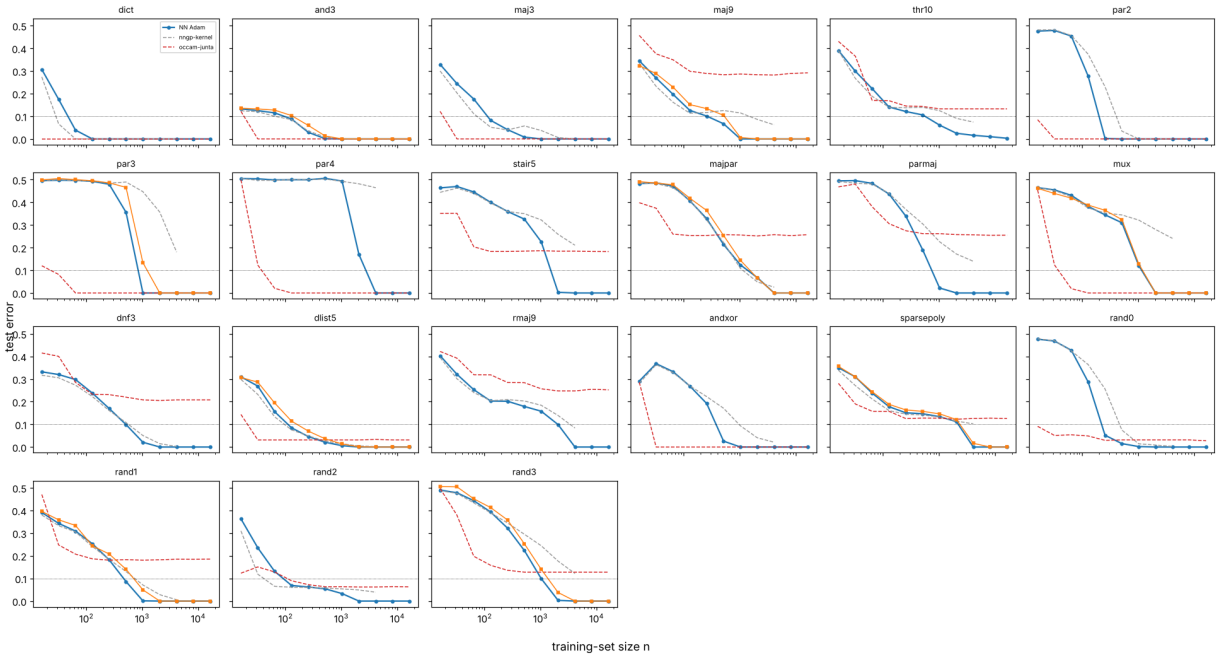


Figure 1: Test error at the end of training against training-set size, $d = 30$. Blue: Adam-trained networks (3 seeds). Orange: μP-SGD-trained networks, on the held-out targets (2 seeds). Grey dashed: NNGP kernel ridge regression. Red dashed: Occam over juntas of up to 4 variables.

3.2 Dependence on the input dimension

Table 2 and Figure 2 show n^* at $d = 15, 30, 60$, and at $d = 120$ for four targets [Numerical]. The growth with d is polynomial, with an exponent that increases with the number of inputs that must be found at once:

- **Parities.** For 2- and 3-bit parities the exponents are 1.04 and 1.91 over $d = 15$–60. From $d = 30$ to 120 the local exponents are 1.09/1.03 for χ_2, on the factor-2 grid, and 1.93/1.96 for χ_3, on a refined grid of n (quarter-octave steps) that gives $n^* = 719$, 2740 and 10691 at $d = 30$, 60 and 120. Both match the leap rate $d^{\max(1,k-1)}$, which on the hypercube matches CSQ lower bounds [3]. The 4-bit parity grows like $d^{2.40}$, below d^3.
- **Leap-1 compositions** (staircase, multiplexer) grow like $d^{1.5}$ over $d = 15$–60, with local exponents falling from 1.7–1.8 to 1.2–1.3. The multiplexer's is 1.20 from $d = 60$ to 120 (factor-2 grid).
- **MAJ$_9$.** Down to test error about 0.1, its learning curve is a function of n/d at all four d, within 0.01–0.04 of a constant-free prediction from the sampling noise of the linear (Hebbian) direction (Figure 2b). Below 0.1 the curves spread (0.02–0.08 at $n/d = 17$). Its apparent n^* exponent comes from where this flat curve crosses 0.1.

Occam-type laws grow like $\log d$, with local exponents of 0.19–0.37 over this range, and fail completely.

Training reuses a fixed sample for many epochs, so the networks are not CSQ learners in the strict sense. In Gaussian single-index models with information exponent k, smoothing the loss lets online SGD reach the CSQ lower bound $d^{k/2}$ [12]. SQ algorithms, or gradient descent that reuses batches, go further, to about d for polynomial links, by exploiting nonlinear functions of the label

target	n^* at $d = 15$ / 30 / 60 / 120	exponent (15–60)	leap d^{k-1}	Gaussian $d^{k/2}$
χ_2	100 / 200 / 427 / 869	1.04	1	1
χ_3	241 / 843 / 3410 / 14188†	1.91	2	1.5
χ_4	508 / 2728 / 14255 / –	2.40	3	2
staircase (5 terms)	433 / 1514 / 3487 / –	1.50	–	–
multiplexer	364 / 1156 / 2848 / 6524†	1.48	–	–
MAJ(x_1x_2, x_3x_4, x_5x_6)	306 / 1380 / 3466 / –	1.75	–	–
MAJ$_9$	123 / 263 / 625 / 845	1.17	–	–

Table 2: Network sample complexity against input dimension (Adam, width 128; 3 seeds at $d = 30$, 2 elsewhere). All entries are on the factor-2 n grid. †A refined (quarter-octave) grid gives 10691 for χ_3 and 5195 for the multiplexer at $d = 120$, and 248 / 719 / 2740 for χ_3 at $d = 15$ / 30 / 60. "–" at $d = 120$: not run; in the rate columns: not applicable. The $d^{k/2}$ column is the CSQ rate for Gaussian single-index models, a reference by analogy. For χ_4 the 20000-step budget limits n^* at both ends of the range, so the sign of its effect on the exponent is unknown.

[13, 14]. For ± 1 labels every statistical query is a correlational query plus a label-free one [9], so that last speed-up has no direct analogue here, and the $d^{k/2}$ column is a Gaussian reference only. For $k \leq 3$ the networks follow the leap exponent $k - 1$; at $k = 4$ they fall between d^{k-1} and $d^{k/2}$.

3.3 Gradient descent against Occam, by leap

Take the 26 targets, designed and random, with at most 4 relevant inputs. The ratio of the networks' n^* to the Occam-over-juntas learner's grows with the target's leap [Numerical]:

leap	1	2	3	4
targets	14	8	2	2
median $n^*_{\mathrm{NN}}/n^*_{\mathrm{Occam}}$ at $d = 30$	8.1	16.6	36	85

Within leap 1 the spread is wide, from 1.7× to 31× (the multiplexer is 31×). The correlations along the target's greedy path matter as well as its leap. Section 4.3 makes this quantitative.

3.4 Robustness: optimizer and width

SGD. μP-SGD on the same training sets gives n^* similar to Adam's on the held-out targets: mean $|\log_2|$ ratio 0.43 [Numerical]. On the 3-parity, SGD needs 382, 1220 and 3562 samples at $d = 15, 30, 60$, against Adam's 241, 843 and 3410. That is an exponent of 1.61 against 1.91. At $d = 60$ both values are interpolated inside the same factor-2 step of the n grid, so each exponent is uncertain by up to about ± 0.25, and the difference is not resolved.

Width. At widths 32, 64, 128 and 256 (2 seeds; 3 at width 128) [Numerical]:

- **3-parity:** n^* does not change visibly (911, 827, 843, 873). At every width the test error is 0.28–0.47 at $n = 512$ and at most 0.04 at $n = 1024$ (exactly 0 at widths 64–256).
- **4-parity:** n^* falls monotonically with width (3554, 3304, 2728, 1899), a factor of 1.9 from width 32 to 256, and has not levelled off.

This pattern does not fit a single width-dependent constant in the noise law of Section 4.3, which would make the 4-parity's width effect the square of the 3-parity's. Nor does it fit a "luckiest unit" account. In that account the largest product of k initial weights over the units would set the escape, and it predicts a width effect for the 3-parity too, since that maximum grows 1.9× from width 32

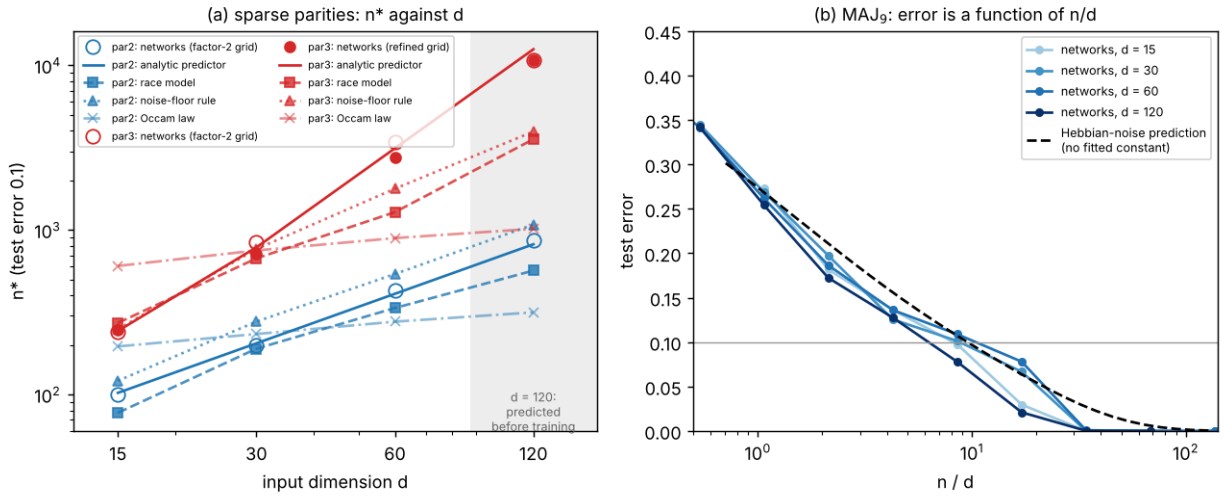


Figure 2: (a) n^* against d for the 2- and 3-bit parities: networks, the analytic noise-law predictor (Section 4.3), the race model, a noise-floor rule, and the Occam law. The predictor's, race model's and Occam law's predictions at $d = 120$ (shaded) were saved before the $d = 120$ networks were trained; the noise-floor rule was evaluated there afterwards. (b) MAJ_9: network test error against n/d at four input dimensions, and the constant-free prediction from the noise of the Hebbian direction.

to 256 at $k = 3$. A guess [Heuristic]: the 3-parity is at its width limit by width 32, its escape carried by the bulk of the units as in mean-field theory, while the 4-parity still relies on lucky units.

4 A discrete model of the feature channel

4.1 The race model

The first model, the *race model*, makes the saddle-to-saddle picture into an algorithm. Its parts:

- **State.** A shared two-level circuit of discrete ReLU gates $g = \mathrm{relu}(\sum_j w_j u_j - b)$. The weights are integers in $\{-3, \ldots, 3\}$, and the wires u_j are input literals and earlier first-level gates. The gates feed a ridge least-squares readout, refitted after every new gate.
- **Proto-units.** A persistent pool of dormant proto-units of orders 1–4 stands in for the not-yet-recruited hidden units.
- **Alignment.** Each proto-unit's direction aligns to the residual by (tensor) power iteration on the training sample. It does so at the speed of the homogeneous law $\dot{m} = kU_k m^{k-1}$, with U_k the order-k utility.
- **Firing.** The rounded gate's utility then drives a clock, and the gate fires when the clock reaches $\theta = \log(1/\alpha) = 1.5$. The earliest gate to fire is added to the circuit, the readout is refitted, and the race resumes.
- **Stopping.** Growth stops at zero training error, or after 48 gates.

Three constants were calibrated on the 12 calibration targets only. The model sees nothing but the training sample.

Fit to the networks [Numerical]. Table 3 lists the scores.

model	mean $\lvert\log_2(n^*_{\text{model}}/n^*_{\text{NN}})\rvert$ held-out (9)	random (20)	$d=15$	$d=60$	per-example	time Sp.
race model	0.33	**0.20**	0.45	0.43	0.50	0.83
race + kernel background (Section 4.4)	0.35	–	0.39	0.66	0.65	0.81
minimal model (2 constants)	0.59	0.47	0.30	0.52	0.50	0.69
analytic predictor (no simulation)	0.38	0.42	0.30	0.52	–	–
sswm (mutate and select)	1.36	1.12	–	–	0.26	0.72
NNGP kernel	2.58	1.78	1.43	2.68	0.69	0.22
Occam over juntas	4.61	3.76	–	–	0.20	−0.07
SGD-trained vs Adam-trained networks	*0.43*	–	–	–	*0.71*	*0.83*
Adam networks from a new initialization	–	–	–	–	*0.62*	–

Table 3: Models scored against the Adam-trained networks. Per-example agreement and the time Spearman are on the held-out targets. The race model's held-out score varies between 0.23 and 0.33 across its internal random streams; its value averaged over streams is 0.28. The hybrid's held-out n^* error is the average over its three streams; its other entries are for one stream (its agreement averaged over streams is 0.66). Its $d=60$ score uses a random-feature background above $n=2048$; with the exact kernel there it is 0.47. "–": not evaluated, or not defined for that model. The $d=15$ agreement of the minimal model and the predictor depends on a cap that was probably added after the $d=15$ network results had been seen, so it is not a clean out-of-sample test. The last two rows compare networks with networks.

- **Held-out targets.** The race model's n^* is within 1.26× of the networks' on average, it ranks the targets identically, and its learning curves match in shape. It reproduces intermediate plateaus. For example, on $\text{sign}(2x_1 + x_2x_3 + x_4x_5x_6)$ test error stays near 1/8 from $n \approx 256$ to $n \approx 2000$, which is the error of the partial predictor $\text{sign}(2x_1 + x_2x_3)$.
- **20 random formulas generated after calibration.** It is within 1.15× on average (Spearman 0.99), better than on the held-out set.
- **Other input dimensions.** At $d=15$ and $d=60$, without recalibration, it is within about 1.35×.
- **The original "mutate and keep improvements" model.** A strong-selection/weak-mutation evolution of a single fan-in-2 circuit does much worse: mean $\lvert\log_2\rvert$ ratio 1.36.

4.2 What makes it work: ablations

Table 4 summarizes 30 ablations, each run on the full grid with the calibrated constants and no recalibration [Numerical]. Every "no change" is judged against the model's own variation across its internal random streams, which spans 0.23–0.33 held-out over six streams.

- **The race dynamics do not set n^* on the held-out targets.** Removing the alignment cost, resetting the clocks after every firing, or replacing the race by plain greedy selection changes held-out n^* error by at most about 0.1 beyond that spread. So does varying θ from 0.5 to 6, the power-iteration steps from 1 to 32, or the pool size from 1 to 32. One exception: removing the alignment cost only for orders ≥ 3 gives 0.50. On the calibration targets, greedy selection (0.40), pool size 1 and one power-iteration step (0.49 each) sit 0.1–0.2 above the band of 0.23–0.30 there, mostly through the 4-parity. So how well the search reaches high-order directions does matter on that side. Compared model against model on the held-out targets, the alignment, clock, greedy and θ variants differ from the six control streams by 0.13–0.21 in $\lvert\log_2 n^*\rvert$, no more than the control streams differ from each other (0.11–0.23). Over all 21

variant	held-out $\lvert\log_2\rvert$ ratio	per-example
race model (calibrated)	0.33	0.50
greedy: add the highest-utility gate, no race	0.27	0.53
one alignment law for all orders	0.28	0.50
depth-1 gates only	0.35	0.51
two other internal random streams	0.32, 0.27	0.51, 0.51
noise-free search (directions from a large fresh sample)	4.63	0.04
independent search sample of the same size n	0.31	0.26
no order-3/4 candidates	1.43	0.52

Table 4: Selected ablations of the race model, held-out targets.

targets, greedy (0.26), pool size 1 (0.32) and one power-iteration step (0.37) move more than a reseed does (0.10–0.21).

- **The noise of the direction search does.** Computing the search directions on a large fresh sample removes their finite-sample noise. n^* then collapses to 16–87 on every target, as far from the networks as the Occam learner (4.63, against its 4.61). Running the whole search (directions, thresholds and utilities) on an independent sample of the *same* size n restores the match (0.31; directions alone: 0.41). So what matters is the noise *level* at sample size n, not overfitting to the particular training labels.
- **The particular sample's noise sets which examples are wrong.** With the independent search sample, per-example agreement falls from 0.50 to 0.26, although n^* is matched. Roughly half of the race model's excess agreement with the networks comes from both being misled by the same spurious correlations of the shared sample [Heuristic].
- **The gate library matters.** Without order-3 and order-4 candidates, the 3- and 4-bit parities, the sparse polynomial and recursive majority are never learned (1.43). Threshold gates in place of ReLU gates score 0.60, and weights in $\{-1, 0, 1\}$ score 0.81. Depth-2 gates help only staircase-like targets, by about 0.4 in $\log_2$.
- **Time.** The ordering of targets by time to generalize survives every ablation, including the noise-free search, so the true features' utilities set it, not the race clocks. Within a target, the time Spearman is about 0.

4.3 The noise law and a closed-form predictor

The ablations suggest a smaller model [Heuristic]: greedy stagewise least squares over ReLU gates with ± 1 weights, in which a candidate is accepted when

$$n\rho^2 \;>\; c\,N(k), \qquad N(k) = d\,\max\!\big(1, \tfrac{d}{a}\big)^{\max(k-2,0)}\ (k \geq 1), \qquad N(0) = |\text{active pool}|. \tag{1}$$

- ρ is the gate's correlation with the current residual.
- k is the number of inputs the gate needs that the circuit does not yet use. Reusing grown features ($k = 0$) costs only a search over the small active pool.
- For $k \geq 3$ the heuristic derivation comes from the alignment law. An order-k unit escapes only if its signal $kUm(0)^{k-1}$ beats the frozen sampling noise $\sim n^{-1/2}$, with an effective initial overlap $m(0)^2 = a/d$. $N(1) = d$ is the sampling noise of the Hebbian unit, and $N(2) = d$ a spectral (BBP-type) threshold. The overlap reading of a is not quantitative.

This is the correlational-SQ / leap scaling $d^{\max(1,k-1)}$. Two constants, $c = 2.25$ and $a = 18$, were calibrated on the calibration targets for the predictor and reused for the minimal model, whose own grid does not resolve its optimum. The predictor's structural choices were made with held-out scores in view.

Evaluated along the target's *population* greedy path, (1) gives a closed form:

$$n^* \approx \max\Big(n_{\text{Hebb}},\ c \max_{i \le i^*} \frac{N(k_i)}{\rho_i^2}\Big).$$

Here i^* is the shortest prefix of the path that reaches error 0.1, and n_{Hebb} is the sampling-noise floor of the linear part. Some closed forms:

- $n^*(x_1) = \Phi^{-1}(0.9)^2 d = 1.64\, d$, with no fitted constant.
- $n^*(\text{MAJ}_9) = 9.52\, d$, with no fitted constant.
- $n^*(\chi_k) = c\,(2^k - 1)\, d \max(1, d/a)^{k-2}$. The first gate, AND_k, has $\rho^2 = 1/(2^k - 1)$ with χ_k, and the remaining gates are cheap $k = 0$ moves.

On the 20 random formulas the analytic predictor, which reads n^* off its predicted learning curve, is within 1.34× on average (Spearman 0.92–0.96, depending on the saved run). The closed form reproduces that curve's n^* for 19 of the 21 designed targets; it was not checked on the random formulas. The simulated minimal model keeps the race model's success agreement and per-example agreement, but not its held-out n^* accuracy (0.59, against an average of 0.28 for the race across its streams).

A pre-registered test at $d = 120$ [Numerical]. The predictor, its variants, and the race model (4 internal streams) were evaluated at $d = 120$ with their $d = 30$ constants. Their predictions were saved before the $d = 120$ networks were launched, and the file timestamps confirm the order.

target	networks	analytic predictor	race model	Occam law
χ_2	869	**817**	566	315
χ_3	10691 (refined grid)	**12525**	3557	1015
MAJ_9	845	1143	1307	1143
multiplexer	5195 (refined grid)	3405	4013	891

The 3-parity is the decisive case.

- **Predictor.** The leap law predicted that n^* would roughly quadruple from $d = 60$ to 120. It did: from 2740 to 10691 on the refined grid.
- **Race model.** From $d = 30$ to 120 it grows only like $d^{1.2}$ (2.8× from 60 to 120), and it is now 3× too optimistic. A possible reading [Heuristic]: its order-k search accepts a direction once it is *detectable* in the empirical moments, while the networks behave as if the direction must be *reached* from a random start. A direct test of reachability from random starts does not explain the 3-parity gap at $d = 60$, however.
- **Why the race model matched at $d = 30$.** At $d = 30$ the two thresholds nearly coincide [Heuristic].
- **Misses of the predictor.** It underpredicts compositions of low-leap moves: the multiplexer by 1.5× at $d = 120$, and by 1.4–2.1× at $d = 30$ and 60. It overpredicts the 4-parity at $d = 60$ by 1.6×.

4.4 Kernel background plus grown features

The race model matches *whether* and *when* the networks generalize, but agrees less with *which* test points they get wrong (0.50) than the NNGP kernel does (0.69). A diagnosis on the partially learned cells (95; most numbers below are on the 44 held-out ones) [Numerical]:

- **The output has two parts.** Split each learner's output into the part that depends only on the target's relevant variables, the signal, and the rest, which comes from the particular training sample. About 45% of a network's output variance is this sample noise: 59% at $n \leq 64$, 23% at $n = 2048$.
- **Every learner gets the signal; the kernel gets the noise.** All learners match the network's signal part (correlation 0.92–0.98). The noise part correlates 0.81 with the kernel's, 0.70 with that of a network retrained from a new initialization, and 0.48 with the race model's.
- **Swapping only the noise patterns** moves the race model's agreement from 0.50 to 0.63, and the kernel's from 0.69 to 0.53.
- **Over training time,** the networks first interpolate within 100–1000 steps, with kernel-like mistakes. Near n^* the generalizing features grow afterwards.
- **The kernel behaves like the network averaged over initializations.** Two single networks trained on the same data from different initializations agree only at 0.62. The average of three such networks agrees with a fourth at 0.684, and this extrapolates to about 0.71–0.73. The kernel agrees 0.78 with a three-network average.

A hybrid model adds an NNGP-kernel background to the race's gates in one joint ridge readout [Numerical]:

- **The background** relaxes on the race clock like kernel gradient flow, with ridge $c_{\mathrm{bg}} n/t$.
- **Growth** stops at a square-loss level, `stop_mse`, after which the background interpolates the remaining residual.
- **Calibration.** Two constants, c_{bg} and `stop_mse`, were chosen on the calibration set, together with the schedule form, the stop rule and the end-of-training step. Held-out agreement was visible while these design choices were explored, so only the constants are strictly calibrated on the calibration set.

Its results, and what they suggest:

- **Agreement.** On the held-out targets it raises per-example agreement from 0.50 to 0.65 (bootstrap over targets: +0.15, interval $[0.13, 0.17]$). The gain holds on every held-out target and every seed.
- **Cost.** Its n^* error is 0.35 against the race model's 0.28, both averaged over random streams.
- **Few grown gates.** It needs about 2 grown gates per partially learned cell, where the race model needs about 22. The race model's extra gates memorize the sample. The networks do that with their random-feature background instead [Heuristic].
- **A simpler competitor does as well on these numbers.** Adding the race's and the kernel's standardized scores, each fitted separately, gives held-out n^* error 0.29 and agreement 0.62. The hybrid's advantage is mechanistic, not numerical: it is one learner, and its background does the memorizing in place of the race's extra gates.

- **A candidate bound.** This suggests an a-priori bound of the form "Occam for a few grown gates, plus a kernel-norm term for the background". We have not attempted it.
- **Remaining discrepancy.** On easy targets the networks are 1.1–2.2× *less* sample-efficient than their own NNGP kernel, for example 47 against 29 samples for x_1. The race model reproduces this through its noisy Hebbian first gate; the hybrid does not.

5 Compositional structure beyond leap

Leap complexity depends only on Fourier supports (Section 7.4). Kaarel Hänni proposed a direct test. With $s = x_1 + x_2 + x_3 + x_4$, consider

$$A = s + x_1x_2 + x_3x_4 + x_1x_2x_3x_4 \qquad \text{and} \qquad B = s + x_1x_2x_3x_4.$$

Both have leap 1: once $x_1, \ldots, x_4$ are found, the degree-4 monomial adds no new input. The prediction was that A should nevertheless be easier, because x_1x_2 and x_3x_4 are learned for their own sake and then combine into $x_1x_2x_3x_4$.

Design.

- **Task.** Real-valued regression with square loss, with every term at coefficient 1/2. A sign-threshold label would wash out the degree-4 term: $\operatorname{sign}(B) = \operatorname{sign}(s + 1)$.
- **Networks.** The same networks as before, at $d = 30$ and $d = 60$.
- **Measure.** n_4^* is the smallest training-set size at which the trained network's test-set Fourier coefficient on $x_1x_2x_3x_4$ reaches 0.4, i.e. 80% of its target.
- **Controls.** We add controls that separate competing explanations:
  - $C = s + x_1x_2 + x_2x_3 + x_1x_2x_3x_4$: in-support pairs that do not multiply to the degree-4 term;
  - $E = x_1x_2 + x_3x_4 + x_1x_2x_3x_4$: no linear part (leap 2);
  - $F = x_1x_2x_3x_4$: leap 4;
  - $G = s + x_5x_6 + x_7x_8 + x_1x_2x_3x_4$: pairs outside the support;
  - $H = s + x_1x_2 + x_1x_2x_3x_4$: one pair.
- **Order.** The predictions of leap complexity, of our move classification, and of the race model were recorded before any network result was inspected, and committed seconds later; the $d = 30$ runs had already started. The race model used its calibrated constants on the networks' exact training sets. For regression, its growth stops once the training MSE is below 2% of the target variance, a new rule that was not calibrated. G, H and the $d = 60$ runs were added after the first $d = 30$ results, with the race model's predictions for them recorded and committed before those runs began.

Results [Numerical]. Table 5 and Figure 3 show the outcome.

1. **The prediction holds, and the effect grows with d.**
   - *Samples.* A needs fewer samples than B: 1.23× fewer at $d = 30$ and 1.59× at $d = 60$.

	target	leap	n_4^*, $d = 30$	n_4^*, $d = 60$	race model, $d = 30$ / 60
E	$x_1x_2 + x_3x_4 + x_1x_2x_3x_4$	2	**1332**	–	1974 / –
C	$s + x_1x_2 + x_2x_3 + x_1x_2x_3x_4$	1	1586	3120	1946 / 7438
A	$s + x_1x_2 + x_3x_4 + x_1x_2x_3x_4$	1	1652	3165	2119 / 2943
H	$s + x_1x_2 + x_1x_2x_3x_4$	1	1842	–	2008 / –
B	$s + x_1x_2x_3x_4$	1	2030	5021	2013 / 26096
G	$s + x_5x_6 + x_7x_8 + x_1x_2x_3x_4$	2	3488	–	1975 / –
F	$x_1x_2x_3x_4$	4	3635	14330	2306 / 15755

Table 5: Composition test: training-set size at which the degree-4 coefficient reaches 80% of its target (seed mean; 3 seeds at $d = 30$, 2 at $d = 60$). The race model's predictions were recorded before the corresponding network results (for G, H and $d = 60$, before the runs). "–": not run.

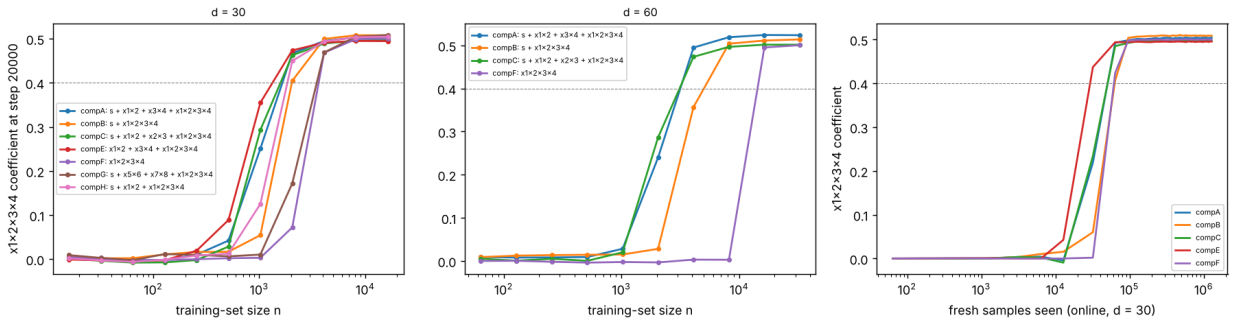


Figure 3: The degree-4 coefficient learned by the networks. Left and middle: fixed training set, at the end of training, against n, at $d = 30$ and $d = 60$. Right: one-pass training with fresh samples, against samples seen.

- *Time.* At $n = 2048$, $d = 30$, the degree-4 term reaches 80% after 2000 steps for A and 20000 for B.
- *One pass.* After 32000 fresh samples the degree-4 coefficient is 0.17–0.25 for A and 0.05–0.07 for B (target 0.5).

Leap complexity gives the two targets the same rate.

2. **Any in-support intermediate feature helps; out-of-support ones do not.**

   - *Non-composing pairs.* C, whose pairs do not multiply to the degree-4 term, is as easy as A, at both $d = 30$ and $d = 60$. Note that $x_1x_2x_3x_4 = (x_1x_2)\,x_3\,x_4$ is still a product of one learned pair and two learned inputs.
   - *One pair.* H lies between A and B.
   - *Pairs on other variables.* G gets no help: its n_4^* is that of the bare parity F, and worse than B's.
   - *Pairs without the linear part.* E is the easiest of all, in both protocols.

   This suggests that what matters is how many already-learned features the degree-4 term needs to combine, not how many new inputs it needs.

3. **Leap gets the asymptotic class partly right.**

   - *Leap-1 targets with pairs.* From $d = 30$ to 60, n_4^* grows like $d^{0.94}$ for A and $d^{0.98}$ for C, as leap 1 predicts.

- *The bare parity.* F grows like $d^{1.98}$, below leap's d^3.
- *B in between.* B grows like $d^{1.31}$. Its degree-4 term is helped by having found $x_1, \ldots, x_4$, but still behaves partly like a higher-order search.

4. **Neither of our predictors anticipated this.**

   - *Our move classification.* Because k counts new inputs, it assigned the degree-4 term in B a cheap $k = 0$ reuse move. Evaluated after the runs, the noise law puts this at a few hundred samples. The four inputs, each with $\rho^2 = 1/5$, become active at $n \approx c\, d/\rho^2 = 2.25 \cdot 30 \cdot 5 \approx 340$, and their AND, with $\rho^2 = 1/15$, then costs $c\,|\text{pool}|/\rho^2 = 2.25 \cdot 4 \cdot 15 \approx 135$. The networks need 2030.
   - *The race model.* Its order-3 and order-4 units search over all d input literals, whether or not they are already active; only its order-1 and order-2 units read grown gates. So it can build A's degree-4 term as a depth-2 product of the two pair gates, while B and C (where $x_1x_2 \cdot x_2x_3 = x_1x_3$) need the higher-order search over literals. At $d = 30$ this made little difference: it gave all targets nearly the same n_4^*. At $d = 60$ it is right for A and F but $5\times$ too pessimistic for B and $2.4\times$ for C.

Reading [Heuristic]. The cost of a high-order term is set by the number of *existing features* that must be combined at once to build it, not by the number of new inputs. For a library of fan-in-2 gates this is the "circuit leap" of Section 7.4: here, combining m features takes $m-1$ gates inserted at once. With gates of fan-in 4 or more, every target here has circuit leap 1.

- In A and E the degree-4 term is a product of two learned pairs.
- In C and H it is a product of a learned pair and two learned inputs.
- In B it is a product of four learned inputs.
- In G the pairs are useless for it, and they compete for the network's capacity and time.

The observed order is $E < C \approx A < H < B < G \approx F$. It is broadly consistent with the reading, but not in detail. A and C tie, although A combines fewer features. E is easier than A, and H harder than C, although each pair combines the same kind of features; their per-seed values do not overlap. $G \gg B$ rests on the competition clause. Explaining these needs extra assumptions, such as a cost of the linear part, that we have not tested.

Consequences for the noise law. The noise law of Section 4.3 should classify moves by the arity of the combination over existing features, with a cost that depends on d only through the inputs that are genuinely new.

- The d-dependence of B ($d^{1.3}$) shows that the noise of such a combination is not d-free.
- The size of the active pool alone does not account for it.

We have not yet built or calibrated this refined law.

6 Generalization bounds

The question here is whether "training has not taken long, so the circuit cannot have grown big, so a Chernoff bound over all small circuits applies" can be made rigorous, and whether it gives useful numbers. Appendices A and B state the results precisely. This section summarizes them.

6.1 What is proved

The Occam step needs no dynamics [Proved]. Consider any learner whose output after T gates is a circuit from a fixed library, followed by a threshold readout. In the realizable case its test error is at most about

$$(T(\log|G| + k\log(d+T)) + T^2\log 2 + \log\tfrac{(T+1)(T+2)}{\delta})/n,$$

simultaneously for all T (Appendix A). The dynamics are needed only to bound T.

Training time does not bound the raw gate count. Many correlated gates can fire at once. On MAJ_m, m^2 gates fire at the same instant of the race. So training time cannot bound the raw gate count uniformly over gate libraries. For a fixed library only a weak time-only bound holds, with a factor of about d. What time does bound, in the race model, are three quantities [Proved]. Time is measured as $\tau(t) = \int_0^t \sqrt{2\hat{L}}$ in units of $\Theta = \log(1/\alpha)$, and the three quantities are:

- the circuit's depth;
- its incoherence-weighted size, $T \le \kappa(\tau/\Theta)^2$, where κ is the top eigenvalue of the fired gates' Gram matrix;
- the readout's ℓ_1 norm, at depth one.

Combined with the Occam step, they give statements of different strength. Besides the weak fixed-library bound, only the norm gives a test-error bound in training time alone, at depth one, and it is near-vacuous. The size bound is a decomposition: κ is a property of the learned circuit, so the bound is never tighter than the Occam bound at the realized size. Depth alone bounds no test error. On targets with no depth-one correlation, such as parities, no gate fires before $t \approx \Theta\sqrt{n/(8\log M)}$. This is a precise version of "it has not had time to memorize yet". That noise gates in general start firing only around this time is [Heuristic].

Diagonal linear networks [Proved]. For a diagonal linear network over a Boolean feature dictionary, small-initialization gradient flow keeps the number of large coordinates at most $\kappa\tau^2/\Theta_\varepsilon^2$. This gives a generalization bound uniform in time, with only ln(dictionary size) dependence.

Two-layer quadratic networks with fixed output signs [Proved]. These are networks $f(x) = \sum_j[(u_j\cdot x)^2 - \|u_j\|^2] - \sum_j[(v_j\cdot x)^2 - \|v_j\|^2] = \langle UU^\top - VV^\top, xx^\top - I\rangle$, with output weights fixed at ± 1 and centred units, equivalently symmetric matrix sensing. Under gradient flow from small initialization, the number of grown hidden units is at most $\kappa(\tau/\Theta)^2$ at every training time.

- κ depends only on the training inputs: not on the labels, the trajectory or the width.
- The bound is weight-free rather than a-priori: τ is read off the training-loss curve. The same statement in raw training time alone is much weaker.

Together with a union bound over low-rank classes, this gives a test-error bound uniform in time. This makes the motivating sketch rigorous for this class of networks, in weight-free form. The bound loosens on plateaus with nonzero residual, where τ keeps growing.

Two-layer ReLU networks (gradient flow from balanced small initialization). We prove three things [Proved]:

- an exact "mass clock" identity;

- that no unit grows before about $\log(1/\alpha)/U^*$, with constant 1, which is sharp;
- a width-free generalization bound valid through the first escape from the initial saddle. It is vacuous below about $n \approx 10^5$ at $d = 16$.

In addition:

- **Merging.** A conserved quantity, the off-diagonal part of the classical balancedness law [4, 15], makes same-sign neurons that share an activation pattern on the training set merge as their mass grows.
- **Beyond the first escape.** Persistence of a small circuit holds under a condition that can be checked from the training residuals [Conditional]. The a-priori statement is open [Conjecture].
- **Reservoir depletion.** In small-α simulations ($\alpha \leq 10^{-3}$, width 64), the first escape recruited every neuron, except one in one run. In one run at $\alpha = 10^{-6}$ the second feature was then not learned by $t = 400$. The discrete models assume the reservoir of dormant units persists; whether it does in the μP networks was not measured.

One-pass training [Proved]. With fresh minibatches, the prequential error certifies the average test error of a window of iterates, with no capacity argument [10]. Taking the window in advance with N fresh samples and M prequential mistakes, the bound is $M/N + \sqrt{\ln(1/\delta)/(2N)}$.

6.2 What can be certified for the trained networks

Parameter-space bounds are vacuous or weak [Numerical]. Norm-based and spectral bounds are vacuous at every $n \leq 16384$, the norm-based ones even at the initialization's norms. Mutual-information bounds are vacuous, and Hardt–Recht–Singer stability does not apply, because its assumptions fail. PAC-Bayes with a prior centred at initialization [16, 23] reaches 0.26–0.56 at $n = 16384$ on the simplest targets only.

Distillation certificates are non-vacuous and never violated [Proved method; Numerical results]. The method:

- Find a small circuit g that agrees with the trained network on *unlabelled* inputs. The candidates are a junta truth table, or the sign of a few Fourier characters with small integer weights.
- Bound the network's test error by an Occam/Chernoff bound for g on the training set, plus an upper confidence bound on the disagreement, estimated on unlabelled inputs not used to choose g.

No labelled hold-out data is used. Over all 21 targets, 5 training-set sizes and 3 seeds:

- 221 of 315 networks get a certificate below 0.5, and 163 one below 0.2.
- No certificate falls below the network's actual test error.
- At $n = 16384$, every network is certified at 0.083 or less, except on three targets, where the certificate's circuit classes do not capture the network cheaply: MAJ of pair parities (0.54), sparse polynomial (0.29) and one random formula (0.42).
- Over training time, the certificates track delayed generalization. On χ_3 at $n = 1024$, the network fits its training set by step 1000 with test error 0.165, and the certificate is 0.23. By step 20000 the test error is 0 and the certificate 0.073. While the network memorizes, the best

certifying class temporarily grows, for example to signs of 8–10 characters, and then collapses to the 3-bit parity.

The race model's own circuits. When the race model's circuit is small, its own-circuit Occam bound is tight: median 0.021 among generalizing fits with at most 6 gates. Half of its generalizing fits carry more than 6 gates, and those bounds are usually vacuous (median 0.67; 34% are below 0.5). Yet most of these large circuits are *functionally* small. A distillation certificate has median 0.125 on a sample of 30 and is below 0.2 in 24 of them. Size of representation and size of function come apart, for the model and for the networks alike.

Online. On the one-pass runs, the prequential bound is within 0.01–0.18 of the test error at the end of each window throughout learning, and about 0.002 after learning. On MAJ_9 it is 0.53, 0.37, 0.20, 0.074, 0.032 and 0.013 over successive windows.

6.3 What the bounds do not answer

The bounds control the gap between training and test error. Once gradient descent generalizes, an Occam bound for the *target's* circuit is already small in every case here: that circuit is tiny and $n \geq n^* \gg k \log d$. The circuits actually found are often large in representation, and their own bounds weaker, as above. The binding question is therefore not the gap but *how many samples until the feature is found*, i.e. the noise law of Section 4.3:

- **Lower half.** For one-pass (CSQ-type) training, its lower half is the known CSQ rate [1, 3]. For training that reuses a fixed sample, no matching lower bound is known, and the 4-parity data scale below d^{k-1}.
- **Upper half.** For real networks, its upper half is open.

7 Discussion

7.1 The picture

Our current picture of rich-regime gradient descent on these targets [Heuristic], summarizing the evidence of Sections 3 and 4:

$$\text{trained network} \approx \underbrace{\text{random-feature kernel background}}_{\text{fits the sample noise; decides which examples are wrong}} + \underbrace{\text{a few grown features}}_{\text{decide whether and when it generalizes}} .$$

Features are added greedily. A candidate is accepted once its correlation with the current residual beats the sampling noise of the *continuous* search that would find it. That noise depends on how many inputs the feature needs at once that the circuit does not yet use, or, for a feature built from grown ones, how many existing features it must combine at once (Section 5); it grows polynomially in d. It is not the best-of-M noise of a union bound over a finite library, so the cost is not a description length.

7.2 Relation to NNBayes ≈ Occam

- **The same answer to *what* is learned.** Once gradient descent generalizes, the network is functionally close to a small circuit, as the distillation certificates show. So does the Occam picture predict [11].

- **Different answers to *how many samples*.**
  - Occam pays the description length, about $k \log d$ for a k-parity. The $\gamma \geq 1$ posterior of [11] is proved to need only polynomially many samples; whether it pays only the description length is open.
  - Gradient descent pays the noise cost of the most expensive move along its greedy path, about d^{k-1} for a move that needs k new inputs at once.
- **The gap grows with leap:** the median ratio is 8×, 17×, 36× and 85× at leaps 1–4, at $d = 30$.

A conjecture suggested by the predictor, not shown here [Conjecture]: the two sample complexities are close for targets with a cheap greedy path, i.e. low-leap compositions with large residual correlations at every step. They differ by about $d^{k-1}/(k \log d)$ when a leap of size k is unavoidable. For fixed k that gap is itself polynomial; it becomes super-polynomial only if k grows with d.

7.3 What is in the literature, and what is new

Known. The qualitative picture is largely in the literature:

- the saddle-to-saddle / incremental-learning view of rich-regime gradient descent [5, 18, 20, 22, 24];
- the leap / CSQ account of which sparse targets online SGD learns, and how fast [2, 3], and its analogue for single-index models, the information exponent [6];
- the equivalence of mutation-and-selection with CSQ learning [17].

New here, as far as we know. The contributions are quantitative and empirical:

1. a calibrated discrete model and a two-constant closed form that predict per-target sample complexity of multi-epoch, fixed-sample training out of sample, including a pre-registered prediction at $d = 120$;
2. ablations showing that the finite-sample noise level of the direction search, not the race dynamics, is the operative ingredient;
3. the kernel-background decomposition of the networks' mistakes;
4. the specific theorems of Appendices A and B: the weight-free rank bound for quadratic networks, and the early-phase results for ReLU networks;
5. the empirical finding that, for networks trained on a fixed sample, distillation certificates are the only tight bounds we found.

7.4 Leap counts supports, not structure

Leap complexity is a function of the target's Fourier *supports* only. Once the inputs of a monomial have been found, any further function of them is free in d. Leap therefore cannot see whether a high-order term can be built from intermediate features that are useful in their own right. Our move classification in (1) inherits part of this blindness, since k counts new inputs. But the correlation ρ of the best available gate depends on which features already exist, so the noise law can in principle see such structure. An early version of this work proposed a "circuit leap" complexity. It orders

a circuit's gates so that each insertion is a positive-utility move, and counts the largest number of gates that must be inserted at once. Section 5 tests the question directly.

7.5 Limitations

- **One setting.** All sample-complexity experiments use one architecture family (a μP two-hidden-layer ReLU MLP), mostly width 128 and Adam, square loss, uniform Boolean inputs, $d \leq 120$, and interactions of at most 4 bits. The theory checks use small two-layer ReLU, quadratic and diagonal linear networks.
- **Width.** The 4-parity's width dependence is not explained by any model here (Section 3.4).
- **The noise law.** Its form for 3 or more simultaneous inputs is a fit, not a derivation. It misses the staircase, multiplexer and MAJ of parities by up to 2.1×, prices the reuse move of the composition test several times too cheaply (Section 5), and overpredicts the 4-parity at $d = 60$.
- **Training budget.** The networks' n^* is still falling at 20000 steps, by about 0.17 in $\log_2$ per doubling of training. Part of every match is a match at this training budget.
- **Verification.** The theorems were checked by AI referees, not by humans (Appendix D).

7.6 Open problems

1. **Prove the noise law** for a two-layer network trained on a fixed sample from small initialization: an order-k feature escapes the plateau once $n \gtrsim d^{\max(1,k-1)}/\rho^2$, up to logarithms, with a matching lower bound for $k \leq 3$. For $k \geq 4$ the networks scale below d^{k-1}, and the right exponent under sample reuse is open. Ingredients exist: information-exponent results for online SGD [6], early alignment [8], and the mass clock and merging law of Appendix B.
2. **Composition costs, and width dependence at order 4.**
3. **Reservoir depletion.** When does the first escape use up the dormant units at practical scales?
4. **A bound of the form "Occam for a few grown features plus a kernel norm for the background"**, as suggested by the hybrid model.
5. **Bayes versus gradient descent.** Characterize the targets on which their sample complexities are close.

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A Bounds for the discrete circuit-growth model and for diagonal linear networks

This appendix states the results of two derivations: Sections A1–A4 of `theory/A-discrete-model.md`, on the discrete circuit-growth model, and Sections C2–C3 of `theory/C-bridge-and-certificates.md`, on diagonal linear networks and a conditional transfer to ReLU networks. Each was refereed twice, and the results are stated as revised after the second round. Numbering in parentheses is the derivation's. Proofs are sketched here; full proofs with explicit constants are in the derivations, and check scripts are in `theory/checks`. "Literature, from memory" marks a published result whose statement the derivation could not check against the paper.

A.1 Setting, counting and the Occam bound

Notation. $\mathcal{X} = \{\pm 1\}^d$, and P is any distribution on $\mathcal{X} \times \{\pm 1\}$, with input marginal P_X. The sample $S = ((x_i, y_i))_{i \le n}$ is iid from P. We write $\langle u, v \rangle_n = \frac{1}{n} \sum_i u(x_i, y_i) v(x_i, y_i)$, $\| \cdot \|_n$ for its norm, and $\bar{u} = \langle u, 1 \rangle_n$ (in population statements, $\bar{u} = \mathbb{E} u$). The test error is $\mathrm{err}(h) = P(h(x) \neq y)$,

and $\widehat{\mathrm{err}}$ is its value on S. Logarithms are natural. $\mathrm{kl}(q\|p)$ is the binary relative entropy, and $\mathrm{kl}_+^{-1}(q,\epsilon) = \sup\{p \geq q : \mathrm{kl}(q\|p) \leq \epsilon\} \leq q + \sqrt{2q\epsilon} + 2\epsilon$. A test-error bound is *useful* if it is at most 0.1, and *non-vacuous* if it is below 1/2.

Circuits. A *library* G is a finite set of gates $\phi : [-1,1]^j \to [-1,1]$ of arity $j \leq k$. It is *Boolean* if every gate maps $\{\pm 1\}^j$ into $\{\pm 1\}$. In a circuit with T gates, gate t applies some $\phi_t \in G$ to sources among the d inputs and gates $1, \ldots, t-1$, and computes g_t. A predictor $h = \psi(g_1, \ldots, g_T)$ has a sign readout $\mathrm{sgn}(c_0 + c \cdot z)$, an output-gate readout $\pm z_T$, or a real readout. $\mathcal{H}_T$ is the set of predictors with T gates. A *learner* is any rule, possibly randomized and data-dependent, that maps S to circuits and readouts. The *toy library* is $\{\mathrm{id}, \mathrm{AND}(\pm a, \pm b), \mathrm{XOR}\}$ up to negation. It has $|G| = 6$ and $k = 2$, and at $d = 30$ it has 2205 depth-one candidates.

Lemma A.1 (Counting (Lemmas 1.1, 1.2 in the derivation))**.** *[Proved] There are at most e^{W_T} circuits with T gates, where $W_T := \sum_{t \leq T} (\log|G| + k\log(d+t-1))$. On N points of $\mathbb{R}^T$, at most $\Phi(N,T) := 2\sum_{i \leq T} \binom{N-1}{i} \leq 2N^T$ functions have the form $\mathrm{sgn}(c_0 + c \cdot z)$. Hence $\log|\mathcal{H}_T| \leq W_T + \rho_T$. Here $\rho_T = \log \Phi(2^{\min(T,d)}, T) \approx T^2 \log 2$ (for $T \leq d$) for a Boolean library with sign readout, $\rho_T = \log \Phi(2^d, T)$ for real-valued gates, and $\rho_T = \log 2$ for an output-gate readout.*

Theorem A.2 (Occam bound, uniform over circuit size (Theorem A1 in the derivation))**.** *[Proved] Fix $\delta \in (0,1)$ and let $L_T := \log|\mathcal{H}_T| + \log \frac{(T+1)(T+2)}{\delta}$. With probability at least $1-\delta$ over S, every $T \geq 0$ and every $h \in \mathcal{H}_T$ satisfy four bounds, where $p = \mathrm{err}(h)$ and $\hat{p} = \widehat{\mathrm{err}}(h)$:*

(1) if $\hat{p} \leq p$, then $\mathrm{kl}(\hat{p}\|p) \leq L_T/n$;
(2) $p \leq \hat{p} + \sqrt{2\hat{p}L_T/n} + 2L_T/n$;
(3) $p \leq \hat{p} + \sqrt{L_T/2n}$;
(4) if $\hat{p} = 0$, then $p \leq 1 - e^{-L_T/n} \leq L_T/n$.

A learner's output with $T(S)$ gates lies in $\mathcal{H}_{T(S)}$, however the gates were chosen. So the bounds hold for any learner, at every stage and every training time, with T the number of distinct gate functions in use.

Proof sketch. For a fixed h, the kl-Chernoff bound gives $P(\hat{p} \leq p,\ \mathrm{kl}(\hat{p}\|p) > \epsilon) \leq e^{-n\epsilon}$. Take a union bound over $\mathcal{H}_T$ with $\epsilon = L_T/n$, and then over T with weights $\delta/((T+1)(T+2))$. Items (2)–(4) follow from $\mathrm{kl}(q\|p) \geq (p-q)^2/2p$, Pinsker's inequality, and $\mathrm{kl}(0\|p) = -\log(1-p)$. Nothing about the dynamics is used. This is the Occam argument [7], made uniform over T. □

Values [Numerical]. Take the toy library at $d = 30$ and $\delta = 0.05$.

- $L_T = 14.8$, 151 and 427 nats at $T = 1$, 10 and 20.
- The realizable bound (4) is useful for $T \leq 2, 7, 20, 62$ and non-vacuous for $T \leq 11, 27, 102, 403$, at $n = 256, 1024, 4096, 16384$.
- The agnostic form (3) is weak: at $T = 5$ and $n = 4096$ its gap is 0.088.
- A symmetrization route in the style of Blumer, Ehrenfeucht, Haussler and Warmuth is larger at every $T \leq 1499$ for $n \in \{1024, 4096, 16384\}$.

A norm version (Theorem A1′, [Proved given three textbook Rademacher lemmas]). This covers $f = c_0 + \sum_j c_j g_j$ with g_j in a finite class $\mathcal{F}$, $\sum_j |c_j| \leq B$ and $|c_0| \leq 1 + B$. Uniformly over $B = 2^j$, with probability $1-\delta$, two gaps are bounded. The clipped square loss exceeds its empirical value by at most $4\mathfrak{R}_n(B) + 2\sqrt{\log(2/\pi_j\delta)/2n}$. The 0-1 error exceeds the empirical ramp loss at a margin θ fixed in advance by at most $\frac{2}{\theta}\mathfrak{R}_n(B) + \sqrt{\log(2/\pi_j\delta)/2n}$. Here $\mathfrak{R}_n(B) = (1+B)/\sqrt{n} + B\sqrt{2\log(2|\mathcal{F}|)/n}$ and $\pi_j = 1/((j+1)(j+2))$.

A.2 The clock race and what training time controls

Definition A.3 (Clock race (Definition 2.1 in the derivation))**.** Fix $\Theta > 0$. The state at time t is a circuit of *fired* gates, a readout $c(t)$, and the residual $r = y - f_t$, where $f_t = c_0 + \sum c_m g_m$.

- A *candidate* m is a gate on inputs or fired gates. Its *availability time* a_m is the last firing time among its gate sources, or 0 if it has none.
- Its *clock* is $S_m(t) = \int_{a_m}^t U_m$, with utility $U_m = \langle r, g_m \rangle_n$. It fires when $|S_m| = \Theta$, with sign $\sigma_m = \operatorname{sgn} S_m$. Several candidates may fire at once, and clocks are never reset.
- After each event the readout is refit, in one of two variants. (LS): least squares with an intercept over all fired gates. (SC), for a depth-one dictionary only: least squares with an intercept over $\mathrm{Sat} = \{m : |S_m| = \Theta\}$, subject to $\sigma_m c_m \geq 0$.

Effective time is $\tau(t) = \int_0^t \|r\|_n = \int_0^t \sqrt{2\hat{L}}$, and $R(t) = \int_0^t r$. The *population race* uses $L^2(P)$ in place of the sample.

Basic facts (Lemma 2.2, [Proved]).

- $|U_m| \leq \|r\|_n \leq 1$, so $\tau \leq t$.
- In (LS), a fired gate keeps $U_m = 0$.
- In (SC), $|S_m| \leq \Theta$.
- With one threshold, rescaling Θ rescales time.

(SC) results hold up to any accumulation point of event times. Whether such points exist is open.

Scope.

- For moves needing $k = 2$ fresh weight layers, $\Theta = c \log(1/\alpha)$, with a constant c that depends on the parametrization ($c = \frac{1}{2}$ for the diagonal network of Appendix A.5).
- Fresh chains with $k \geq 3$ cost about $\alpha^{-(k-2)}$, so the first move of a three-layer network started at scale α costs about α^{-1} [Heuristic]. The depth-one race therefore models diagonal or 2-homogeneous networks.
- The diagonal network's small-α limit [24] is recalled to match (SC) up to the intercept. AGF [20] also accumulates utility without resets. Both are Literature, from memory.
- (LS) is not the depth-one limit.
- Deeper moves are [Heuristic].

Proposition A.4 (Depth (Proposition A2.1 in the derivation))**.** *[Proved] In either variant, a gate of depth D fires only when $\tau(t) \geq D\Theta$. So the depth at time t is at most $\lfloor \tau(t)/\Theta \rfloor \leq \lfloor t/\Theta \rfloor$. The proof is by induction on D, since $\Theta = |S_m(t_m)| \leq \tau(t_m) - \tau(a_m)$.*

Proposition A.5 (No uniform gate-count bound (Proposition A2.2 in the derivation))**.** *[Proved] Consider the population race with uniform inputs and $y = \mathrm{MAJ}(x_1, \ldots, x_m)$, for odd $m \leq d$. Let $u_m = \binom{m-1}{(m-1)/2} 2^{1-m} \approx \sqrt{2/\pi m}$.*

(1) For $G = \{\mathrm{id}\}$, m gates fire together at Θ/u_m and nothing fires earlier, so $T \approx \frac{2}{\pi}(t/\Theta)^2$.

(2) For the toy library, exactly m^2 functions, distinct up to sign, fire together at Θ/u_m: the identities, $\mathrm{AND}(x_i, x_j)$ and $\mathrm{AND}(\neg x_i, \neg x_j)$. So $T \approx (\frac{2}{\pi})^2 (t/\Theta)^4$.

(3) Duplicates with equal availability times fire together, so no function of t/Θ bounds the syntactic gate count uniformly over libraries.

Proof sketch. $\mathbb{E}[yx_i] = u_m$, the chance that the other votes tie, and $\mathbb{E}[yx_ix_j] = 0$. Expanding the AND gates as $\mathrm{AND}(\epsilon_1 a, \epsilon_2 b) = (\epsilon_1 a + \epsilon_2 b + \epsilon_1\epsilon_2 ab - 1)/2$ shows that the listed gates have utility $\pm u_m$, and all others at most $u_m/2$ in absolute value. □

So no bound $T \le C(t/\Theta)^2$ with C independent of d holds for the toy library uniformly over the targets MAJ_m, $m \le d$. At $n = 4096$, maj3 fires 9 gates at $t/\Theta \in [1.98, 2.12]$ [Numerical].

Theorem A.6 (Gate count times incoherence (Theorem A2.3 in the derivation)). *[Proved] Let K_B be the centred empirical Gram matrix of a set B of gates, with entries $\langle g_m - \bar{g}_m, g_{m'} - \bar{g}_{m'} \rangle_n$, and $\kappa_B = \lambda_{\max}(K_B)$.*

(1) In (LS), at any depth, every set B fired by time t satisfies $|B| \le \kappa_B(\tau(t)/\Theta)^2$. If all $a_m = 0$, τ may be replaced by $\|R(t)\|_n$.

(2) In (SC), $|\mathrm{Sat}(t)| \le \kappa_{\mathrm{Sat}(t)}(\|R(t)\|_n/\Theta)^2$, and $\operatorname{supp} c \subseteq \mathrm{Sat}$.

So the number $T(t)$ of distinct fired functions is at most $\kappa(t)(\tau/\Theta)^2$. Let μ_B be the largest absolute centred covariance within B, and $x = (\tau/\Theta)^2$. If $\mu_B x < 1$, then $|B| \le (1 - \mu_B)x/(1 - \mu_B x)$, by Gershgorin's theorem.

Proof sketch. Each $m \in B$ has $\sigma_m S_m(t) = \Theta$. The refit intercept gives $\bar{r} = 0$, so $U_m = \langle r, \tilde{g}_m \rangle_n$ with $\tilde{g}_m = g_m - \bar{g}_m$. With $B(s) = \{m \in B : a_m \le s\}$, $|B|\Theta = \int_0^t \langle r, \sum_{B(s)} \sigma_m \tilde{g}_m \rangle_n \le \tau(t)\sqrt{\kappa_B |B|}$. The last step uses the fact that a principal submatrix has no larger top eigenvalue. When all $a_m = 0$, apply Cauchy–Schwarz to $\langle R, \cdot \rangle_n$ instead. □

The population analogue is tight up to a factor $\pi/2$ (Proposition A.5(1)). A library-wide coherence does not help, since the toy library has $\mu = \frac{1}{2}$.

Theorem A.7 (ℓ_1 norm of the readout (Theorem A2.4 in the derivation)). *[Proved] In (SC) on a depth-one dictionary, $\Theta\|c(t)\|_1 = \langle R(t), f_t \rangle_n \le \|R(t)\|_n \sqrt{1 - \bar{y}^2} \le \tau(t)$, and $|c_0| \le 1 + \|c\|_1$.*

Proof sketch. KKT gives $c_m S_m = |c_m|\Theta$, with $S_m = \langle R, g_m \rangle_n$. Since $\langle R, 1 \rangle_n = 0$, $\Theta\|c\|_1 = \langle R, f_t - \bar{y} \rangle_n$. Here $f_t - \bar{y}$ is the projection of $y - \bar{y}$ onto a cone, so its norm is at most $\sqrt{1 - \bar{y}^2}$. This is LASSO duality, with R as the residual and Θ as the penalty. □

Checks [Numerical]. These are toy races at $d = 30$ with $n = 64$–4096.

- Theorems A.2 and A.6 and Proposition A.4 held at every event.
- The ratio $\kappa\tau^2/(\Theta^2 T)$ is 1.00 on par2, where the bound is attained. It is at most 7 on maj3, and3 and dnf3, and up to 400 on par3, where one noise gate fires late.
- Theorem A.7 holds to machine precision in the toy (SC) runs, and to $3 \cdot 10^{-14}$ on 40 random correlated dictionaries with 130 exits from the saturated set.

Corollary A.8 (Test error along training (Corollary A2.5 in the derivation)). *[Proved for (a), (a′); proved given the lemmas of A1′ for (b)] Let $h_t = \operatorname{sgn} f_t$ and $D(t) = \lfloor \tau(t)/\Theta \rfloor$. Each of (a), (a′) and (b) holds with probability $1 - \delta$, simultaneously for all t.*

(a) For (LS), Theorem A.2 holds for h_t with $T(t) \le \lfloor \kappa(t)\tau^2/\Theta^2 \rfloor$. For example, at zero training error, $\mathrm{err}(h_t) \le L_{\lfloor \kappa\tau^2/\Theta^2 \rfloor}/n$.

(a′) For (LS) or (SC), in a race that merges candidates equal up to sign on the sample: let $\mathcal{F}_{\le D}$ be the non-constant depth-$\le D$ functions that are distinct up to sign on the sample, and $\bar{\kappa}_D$ their top centred Gram eigenvalue. These depend only on the library and the unlabelled inputs. Then Theorem A.2 holds with $T(t) \le \min\{|\mathcal{F}_{\le D(t)}|, \lfloor \bar{\kappa}_{D(t)}\tau^2/\Theta^2 \rfloor\}$, and at depth one τ may be replaced by $\|R\|_n$.

(b) For (SC) at depth one, Theorem A1′ holds with $B \le 2\max(1, \tau/\Theta)$. Its gap is therefore of order $(1 + \tau/\Theta)\sqrt{\log(|G|d^k)/n}$, whatever the gate count.

For (a′), the fired representatives lie in $\mathcal{F}_{\le D(t)}$ (Proposition A.4), and up to sign flips their Gram matrix is a principal submatrix of the library's.

How strong these are.

- *(a) is a decomposition, not a time-only bound.* $\kappa(t)$ belongs to the learned circuit and can grow (3 for maj3 after its first event). It is never tighter than Theorem A.2 at the realized size.
- *(a′) is time-only but weak.* For the toy library, $\bar{\kappa}_1 = d$ in the population, and 35 and 32 empirically at $n = 4096$ and 16384. At zero training error, (a′) gives 0.21 and 0.053 at $\|R\|_n = \Theta$, and 0.61 and 0.195 just below 2Θ. Theorem A.2 at par2's realized size gives 0.004 and 0.001.
- *(a′) needs the merging.* A referee counterexample shows that it fails without it.
- *(b) is near-vacuous.* Its 0-1 bound at $n = 4096$ is 0.40–0.97.

So training time bounds the depth, the incoherence-weighted size and the depth-one readout norm. It bounds the raw gate count only through $\kappa(t)$ or $\bar{\kappa} \approx d$.

A.3 Minibatches: reproducing the population run

The population greedy run (PG). PG adds one gate per step. Given C_t, let f_t be the $L^2(P)$ projection of y onto $V_t = \mathrm{span}\{1, C_t\}$, and $r_t = y - f_t$. PG adds $m_t = \arg\max_m |U_t(m)|$, where $U_t(m) = \mathbb{E}[r_t g_m]$. The maximum is over $M_t \le |G|(d+t)^k$ candidates, identified up to sign. Three assumptions are made:

- a gap $\gamma > 0$: $|U_t(m_t)| \ge |U_t(m)| + \gamma$ for $m \neq \pm m_t$ and $t < T$;
- $|r_t| \le R$;
- $\lambda_* = \min_{t \le T} \lambda_{\min}(K_t) > 0$, for the $L^2(P)$ Gram matrices K_t of $(1, C_t)$.

The empirical run (EG). At step t, EG uses a batch B_t of size b: either S itself (full batch) or a fresh batch. It refits on B_t, adds a maximizer of $|\langle \hat{r}_t, g_m \rangle_{B_t}|$, and outputs the sign of its final refit.

Theorem A.9 (Decision stability (Theorem A3.1 in the derivation))**.** *[Proved] Let* $\epsilon = \gamma/(2 + 2\sqrt{2T/\lambda_*})$*, and suppose that* $b \ge \frac{2R^2}{\epsilon^2} \log \frac{4\sum_{t<T}(M_t+t+1)}{\delta}$ *and* $b \ge \frac{8(T+1)^2}{\lambda_*^2} \log \frac{4(T+1)^3}{\delta}$*. Then, for both protocols, with probability* $1 - \delta$*:*

(1) EG's circuit computes C_T*;*

(2) if $y \in V_T$ *almost surely, EG's predictor equals* y *almost surely.*

With probability $1 - 2\delta$*, Theorem A.2 holds for EG's predictor, on the sample of its final refit, with* $L = \rho_T + \log(1/\delta)$ *and no wiring term.*

Proof sketch. The functions r_t, g_m and K_t belong to the population run, so Hoeffding's inequality and a union bound give ϵ-accurate empirical utilities, and $\lambda_{\min}(\hat{K}_t) \ge \lambda_*/2$. Inductively, if EG holds C_t, then $\hat{r}_t = (I - \hat{\Pi}_t) r_t$ with $\|\hat{\Pi}_t r_t\|^2 < 2(t+1)\epsilon^2/\lambda_*$. So every empirical utility is within $\gamma/2$ of U_t, and the gap fixes the argmax. □

Related results.

- *Reuse* (Theorem A3.5, [Proved]). If batches are drawn with replacement from a fixed S of size n, the same conclusions hold, with the Occam bound on S. The conditions use $(n^{-1/2} + b^{-1/2})^{-2}$ in place of b, and $\delta/2$ in place of δ.
- *Thresholds* ([Numerical]). For par2, with XOR $\in G$, $\gamma = \frac{1}{2}$, $R = \lambda_* = 1$ and $M_0 = 2205$, the theorem needs $b \ge 2254$. With reuse at $n = 16384$ it needs $b \ge 6225$. So the experiments' $b = 64$ is not covered. At $T = 30$ and $\lambda_* = \frac{1}{2}$, the Gram condition needs $4.5 \cdot 10^5$ samples.

- *Limits.* Stability gives exact recovery with no factor $1/\varepsilon$ for a target error ε, but it pays $1/\gamma^2$ and says nothing under ties or zero utilities.
- *The SGD race.* For the race, only clock concentration is proved (Lemma A3.2, [Proved]). Under fresh batches, with residuals bounded by R and a fixed finite candidate set $\mathcal{M}$, with probability $1-\delta$ every clock stays within $R\sqrt{2\eta t_0 \log(2|\mathcal{M}|/\delta)/b}$ of its drift clock up to time t_0; the observed errors are 75–88% of this [Numerical]. That the race reproduces the population race is [Heuristic].
- *Online-to-batch* (Proposition A3.3, [Proved]; cf. [10]). In one-pass training on fresh samples, pre-update minibatch losses estimate the trajectory's average test loss.

A.4 Greedy growth versus minimum circuits

Proposition A.10 (Equal, overshoot, never (Propositions A4.1–A4.3 in the derivation))**.** *[Proved; item 2 for the population race only] Let s^* be the size of the smallest consistent circuit.*

(1) If $y = \pm g^*$ for a depth-one candidate g^*, and $0 < P(y=1) < 1$, then g^* and its duplicates fire first, at $\Theta/\mathrm{Var}(y)$. The race then stops with $T = 1 = s^*$. The same holds on a sample containing both labels, with Var_n, and with duplicates meaning candidates equal to $\pm y$ on the sample.

(2) Take uniform inputs, the toy library, and $y = \mathrm{maj}_3$. Then 9 gates fire at 2Θ, and the 0-1 error is already zero. Exactly 12 depth-two gates of utility $\frac{1}{4}$ fire next, at 6Θ. So the race ends with 21 gates, while $s^* = 3$ ([Numerical], by exhaustive search). For MAJ_m, m^2 gates fire where m suffice.

(3) Take uniform inputs and $y = \chi_A$ with $|A| = s > k$. Then nothing ever fires in the population race. On a sample, with probability $1-\delta$, nothing fires before $\Theta\sqrt{n/(8\log(2(M_1+1)/\delta))}$, where M_1 is the number of depth-one candidates (Proposition A2.6). If $\mathrm{XOR} \in G$, then with probability $1-\delta$ the minimum-consistent-circuit learner with an output-gate readout has error at most $L_{s-1}/n = O((s\log(d+s) + \log\frac{1}{\delta})/n)$.

Proof sketch. (1) For $|g| \le 1$, $|\mathbb{E}[(y-\bar{y})g]| \le \mathbb{E}|y-\bar{y}| = \mathrm{Var}(y)$, with equality only for $g = \pm y$. (2) The 9 gates span the degree-≤ 2 functions, which leaves the residual $-\frac{1}{2}x_1x_2x_3$, and a Fourier lemma bounds the depth-two utilities by $\frac{1}{4}$. (3) Depth-one gates see fewer than s variables, and the error bound is Theorem A.2(4). □

So both learners get the same Occam bound as a function of T, and they differ in which T they reach [Numerical].

- maj3 reaches zero training error with 6 gates at $n \ge 256$.
- dnf3 needs 18 gates where 6 suffice. Its bound at $n = 4096$ is 0.084, where 6 gates would give 0.019.
- par3 fires only noise gates, at $t/\Theta = 2.65$–20.0 (the proved lower bound is 0.84–6.71). Its bound of 0.52–0.95 certifies nothing.

A.5 Diagonal linear networks (Section C2 of the derivation)

Setting. The dictionary consists of D characters $\chi_J(x) = \prod_{i\in J} x_i$; for example, taking all $|J| \le k$ gives $D = D_k = \sum_{i\le k}\binom{d}{i}$. Let $X_{iJ} = \chi_J(x_i)$ and $K = X^\top X/n$. The model is $f_w = \sum_J w_J\chi_J$, with $w = u\odot u - v\odot v$ and loss $L(w) = \frac{1}{2n}\|Xw - y\|^2$. It is trained by gradient flow from $u = v = \alpha\mathbf{1}$. Write $r = y - Xw$, $g = \nabla L = -X^\top r/n$, $G = \int_0^t g$, $\tau = \int_0^t \|r\|_n$ and $R = \int_0^t r$, and set $\Theta_\epsilon = \frac{1}{4}\operatorname{arsinh}\frac{\epsilon}{2\alpha^2} \ge \frac{1}{4}\log\frac{\epsilon}{\alpha^2}$.

Closed form and no early growth (Lemma C2.1, Theorem C2.2, [Proved]).

- $w(t) = -2\alpha^2 \sinh(4G(t))$. So $-G_J$ is exactly the race clock, and $|w_J| \geq \epsilon$ exactly when it reaches $\Theta_\epsilon \approx \frac{1}{2}\log\frac{1}{\alpha}$.
- Since $|G_J| \leq \tau$, nothing reaches ϵ while $\tau(t) < \Theta_\epsilon$.
- Let $U_0 = \max_J |\langle y, \chi_J\rangle_n|$ and $\epsilon_D = \sqrt{\lambda_{\max}(K)D}\,\epsilon$. If $U_0 > \epsilon_D$, the first escape time lies in $[\Theta_\epsilon/(U_0+\epsilon_D), \Theta_\epsilon/(U_0-\epsilon_D)]$.

Theorem A.11 (How many coordinates can have grown (Theorem C2.3 in the derivation))**.** *[Proved] Let $B = \{J : |w_J(t)| \geq \epsilon\}$, $N_\epsilon = |B|$ and $\kappa_B = \lambda_{\max}(X_B^\top X_B/n)$.*

(1) $N_\epsilon \leq (\int_0^t \|\nabla L\|_2)^2/\Theta_\epsilon^2$.

(2) $N_\epsilon \leq \kappa_B\tau^2/\Theta_\epsilon^2$. Let $\mu_n = \max_{J\neq J'} |\langle \chi_J, \chi_{J'}\rangle_n|$ and $x = (\tau/\Theta_\epsilon)^2$. Then $N_\epsilon \leq (1-\mu_n)x/(1-\mu_n x)$ when $\mu_n x < 1$. For uniform inputs and characters of degree at most k, with probability $1-\delta'$, $\mu_n \leq \bar{\mu} := \sqrt{2\log(2D_{\min(2k,d)}/\delta')/n}$.

(3) $\sum_J |w_J||G_J| = \langle R, f_w\rangle_n$, so $\sum_{J\in B}|w_J| \leq 2\tau/\Theta_\epsilon$.

Proof sketch. Each $J \in B$ has $|G_J| \geq \Theta_\epsilon$, and (1) follows from Minkowski's inequality. For (2), with $\sigma_J = -\operatorname{sgn} G_J$, $N\Theta_\epsilon \leq \int_0^t \langle r, \sum_B \sigma_J\chi_J\rangle_n \leq \tau\sqrt{\kappa_B N}$, as in Theorem A.6. The bound on μ_n is Hoeffding's inequality for the characters $\chi_{J\triangle J'}$. □

Item (2) is exact at the first crossing for a single-character target that is empirically orthogonal to the rest of the dictionary. At nonzero coherence it is strict; the derivation sketches, without a full proof, that it becomes tight as $\alpha \to 0$.

Theorem A.12 (Generalization uniform in t and ϵ (Theorem C2.4 in the derivation))**.** *[Proved] Let $\ell_N = \log\binom{D}{N} + \log\Phi(2^{\min(N,d+1)}, N)$ and $\mu_\epsilon(t) = \sum_{J\notin B}|w_J(t)|$. With probability $1-\delta$, for all $t \geq 0$ and $\epsilon > 0$, with $N = N_\epsilon(t)$:*

$$P(yf_{w(t)} \leq 0) \leq \mathrm{kl}_+^{-1}(\widehat{P}_S[yf_{w(t)} \leq 2\mu_\epsilon(t)],\ (\ell_N + \log\tfrac{(N+1)(N+2)}{\delta})/n).$$

Proof sketch. Split $f_w = f_L + f_s$, with f_L supported on B and $\|f_s\|_\infty \leq \mu$. The set $\{yf_L \leq \mu\}$ is a threshold of $(y\chi_J(x))_{J\in B}$, which takes at most $2^{\min(N,d+1)}$ values. By Cover's count there are therefore at most e^{ℓ_N} such sets. Apply kl-Chernoff with a union bound over the sets and over N, and compare margins on test and training points. □

Corollary A.13 (Time-indexed form (Corollary C2.5 in the derivation))**.** *[Proved] Fix $\gamma \in (0,1]$. Let $\epsilon = \gamma/2D$, $\Theta_\gamma = \frac{1}{4}\operatorname{arsinh}\frac{\gamma}{4D\alpha^2}$, $x = (\tau/\Theta_\gamma)^2$ and $\ell_N^* = \max_{N'\leq N}\ell_{N'}$. On the event of Theorem A.12, every $\bar{N} \in [N_\epsilon(t), D]$, possibly data-dependent, satisfies*

$$\operatorname{err}(\operatorname{sgn} f_{w(t)}) \leq \mathrm{kl}_+^{-1}(\widehat{P}_S[yf_{w(t)} \leq \gamma],\ (\ell_{\bar{N}}^* + \log\tfrac{(\bar{N}+1)(\bar{N}+2)}{\delta})/n).$$

There are three admissible choices of $\bar{N}$, each capped at D:

(1) $\lfloor\kappa_B x\rfloor$, with κ_B measured;

(2) $\lfloor(1-\mu_n)x/(1-\mu_n x)\rfloor$, if $\mu_n x < 1$;

(3) for uniform inputs and all characters of degree $\leq k$, the same with $\bar{\mu}$ in place of μ_n, and optionally t in place of τ, if $\bar{\mu}x < 1$, at total probability $1-\delta-\delta'$.

Only choice (3) depends on (n, D, τ) alone, and it needs $n > 2\log(2D_{\min(2k,d)}/\delta')(\tau/\Theta_\gamma)^4$. D enters only through $\log D$.

Discrete GD and SGD (Proposition C2.6, [Proved]). Suppose $2\eta\bar{G} < 1$, where $\bar{G} = \max_j \|g_j\|_\infty$.

- No coordinate grows before the minibatch effective time $\eta \sum_j \|r_j\|_{B_j}$ reaches $\Theta_\epsilon(1 - 4\eta^2\bar{G}^2)$.
- Under SGD, the count is bounded only through the integrated gradient norms $\eta \sum_j \|g_j\|_2$, which is loose.
- Under full-batch GD, the $\kappa_B\tau^2$ form holds, with Θ_ϵ reduced by an explicit $O(\eta^2\bar{G}^2\tau)$.

Checks [Numerical]. These are at $d = 10$, $k = 3$, $D = 176$, with exact test errors. The generalization bounds are at $t = 60$ and $\gamma = \frac{1}{2}$.

- *Size bounds.* The worst ratio of N to $\kappa_B\tau^2/\Theta^2$ is 0.33–0.94, and 1.00 for par2 at $\alpha = 10^{-8}$. The discrete counts reach 0.62–0.80 of their bound under GD, and only 0.07–0.12 under SGD.
- *A posteriori.* At $\alpha = 10^{-3}$, Theorem A.12 gives 0.107 for par2 ($n = 100$), 0.275 and 0.077 for maj3 ($n = 100$ and 400), and 0.166 for and3 ($n = 400$).
- *Time-indexed.* Choice (1) gives 0.218, 0.810, 0.316 and 0.477 on the same runs. Choices (2) and (3) are vacuous except for par2 at $\alpha = 10^{-8}$, where they give 0.107 and 0.163. So the time-only form works here only for a single character at tiny α.
- *Off-dictionary targets.* maj5 lies outside the dictionary and gets 0.951, against a true error of 0.073. par4 is uncorrelated with the dictionary and gets 0.988.

A.6 Conditional transfer to ReLU networks (Section C3 of the derivation)

Known results and growth lemmas. Rigorous results cover early alignment and the first escape in two-layer and homogeneous networks, for example [8]. Exact saddle-to-saddle limits are known only for diagonal networks [20, 24] (Literature). The derivation proves one-sided growth bounds (Lemmas C3.1, C3.2, [Proved]). Let λ_X be the top eigenvalue of the second moment of $(x, 1)$.

- *Two layers.* In a two-layer ReLU network under gradient flow or descent, each unit's $\rho_j = a_j^2 + \|w_j\|^2$ satisfies $\rho_j(t) \le \rho_j(0)e^{2\sqrt{\lambda_X}\tau(t)}$, with τ the discrete effective time $\eta \sum_k \|r_k\|_n$ for descent.
- *Deep networks.* In a bias-free deep ReLU network with $L \ge 3$ under gradient flow, all layers grow equally [15]. The quantity $Q = \max_l \|W_l(0)\|_F^2 + \Delta$ satisfies $Q(t)^{1-L/2} \ge Q(0)^{1-L/2} - (L-2)\sqrt{\lambda_X}\tau$. So, with layers of Frobenius norm $\asymp \alpha$, escape takes effective time at least of order $\alpha^{-(L-2)}$.
- *Tightness* ([Numerical]). At $L = 3$ this is loose by 14–21×, and at $L = 4$ it is vacuous.
- *Scope.* The study's μP Adam networks are outside both lemmas.

The hypothesis. Let $f_{\theta(t)}$ be the network at checkpoints $t \in \mathcal{T}$. Run the (LS) race on the same S with a finite library. Let $h_s^M = \operatorname{sgn} f_s^M$ be its state at race time s, with $T^M(s)$ gates, and let $s(t)$ be a clock calibration. The *saddle-to-saddle approximation hypothesis* (SSA) has two forms:

- sign form: $P_X(\operatorname{sgn} f_{\theta(t)} \ne h^M_{s(t)}) \le \eta(t)$;
- L^2 form: $\|\operatorname{clip} f_{\theta(t)} - \operatorname{clip} f^M_{s(t)}\|_{L^2(P_X)} \le \eta(t)$.

For a Boolean library, let $L_T^{\mathrm{sgn}} = W_T + \log \Phi(2^{\min(T,d)}, T)$ and $L_T^{\mathrm{mar}} = W_T + \log \Phi(2^{\min(T+1,d+1)}, T+1)$. For real-valued gates the exponents are d and $d+1$.

Theorem A.14 (The network inherits the race bound (Theorem C3.4 in the derivation))**.** *[Conditional: proved under SSA] There is an event of probability $1-\delta$ over S, not depending on the network, on which the following hold for every $t \in \mathcal{T}$. Here $s = s(t)$ and $T = T^M(s) \le \kappa^M(s)(\tau^M(s)/\Theta)^2$.*

(1) Under the sign form, $\operatorname{err}(\operatorname{sgn} f_{\theta(t)}) \le \eta(t) + \mathrm{kl}_+^{-1}(\widehat{\operatorname{err}}_S(h_s^M), (L_T^{\mathrm{sgn}} + \log \frac{2(T+1)(T+2)}{\delta})/n)$.

(2) Under the L^2 form, for every $\gamma \in (0,1)$, $\mathrm{err}(\mathrm{sgn} f_{\theta(t)}) \le \frac{\eta(t)^2}{\gamma^2} + \mathrm{kl}_+^{-1}(\widehat{P}_S[y f_s^M \le \gamma], (L_T^{\mathrm{mar}} + \log \frac{2(T+1)(T+2)}{\delta})/n)$.

Proof sketch. Before training, fix two Occam events, each at level $\delta/2$: one for threshold-readout circuits, and one for margin sets of circuits. The sign form then adds the disagreement. In the L^2 form, $y f_\theta \le 0$ forces either $y f^M \le \gamma$ or a clipped gap larger than γ, and Chebyshev's inequality bounds the latter. □

What this gives.

- *A conversion only.* The theorem converts the hypothesis into a bound, and its size term is data-dependent through κ^M.
- *Testing SSA* (Corollary C3.5, [Proved]). Held-out unlabelled inputs make the sign form checkable. The distillation certificate (Theorem B.5) then bounds the network with no hypothesis at all. So SSA is needed only for the predictive statement.
- *A first test* ([Numerical]). On the saved μP Adam networks, and3 and dnf3 match the race's constant state on their first plateau, which is trivial. In transitions, the best disagreement with any race state is 0.1–0.3 (dnf3, mux, stair5), or no better than the disagreement with the target (maj3, majpar, parmaj). On par3 the race never starts, while the network learns.

So Theorem A.14 cannot yet be applied to the study's runs, and proving SSA beyond the first escape is open.

B Bounds for the trained networks

This appendix collects the generalization bounds proved for actual networks rather than for the discrete race model. The sources are three derivations: `theory/B-nn-time-bounds.md` (parameter-space bounds for the study's networks), `theory/C-bridge-and-certificates.md` (distillation certificates and the online protocol) and `theory/D-early-phase-theorem.md` (early-phase theorems for two-layer networks). Each was refereed twice, the second pass finding only minor issues (reports in `theory/checks/`). Results keep the derivation's numbering in parentheses. [Proved] means a complete proof with explicit constants is in the derivation, possibly invoking a named published result; [Conditional] means proved under a stated, unproved hypothesis; [Numerical] means a proved bound evaluated by a script. Sketches give only the key idea. Appendices B.1 to B.5 concern the study's networks. Appendices B.6 and B.7 concern two-layer networks trained from small initialization, and none of their theorems covers the study's μP Adam networks.

Notation. Inputs are uniform on $\{\pm 1\}^d$, labels are $y = f^\star(x) \in \{\pm 1\}$, and $S = ((x_i, y_i))_{i \le n}$ is iid. $\mathrm{err}(h) = \Pr(h(x) \ne y)$ and $\mathrm{dis}(h, g) = \Pr_x(h(x) \ne g(x))$; hats denote empirical versions on S or on an unlabelled sample U, and $\widehat{P}_S$ is an empirical frequency. $\langle u, v \rangle_n = \frac{1}{n}\sum_i u(x_i)v(x_i)$ and $\|u\|_n^2 = \langle u, u \rangle_n$. For real f, $\mathrm{err}(\mathrm{sgn} f) \le \Pr(yf \le 0)$, which the margin results bound. With kl the binary relative entropy, $\mathrm{kl}_+^{-1}(q, c) = \sup\{p \in [q, 1] : \mathrm{kl}(q\|p) \le c\} \le q + \sqrt{2qc} + 2c$. The study's network is $f_\theta(x) = a \cdot \sigma(W_2 \sigma(W_1 x + b_1) + b_2)$, with $\sigma = \mathrm{relu}$, $d = 30$ and widths $N = 128$. Its parameter groups $g = 1, 2, 3$ are (W_1, b_1), (W_2, b_2) and a, with p_g entries. θ_0 is the μP initialization, independent of S. $L_S(\theta) = \frac{1}{2n}\sum_i (f_\theta(x_i) - y_i)^2$, $L_0 = L_S(\theta_0) \approx 0.5$, and $R(\theta) = \Pr[y f_\theta(x) \le 0]$. Every update is $\theta_{k+1} = \theta_k - \eta M u_k$ with per-group multipliers $M = \mathrm{diag}(m_g)$: SGD ($\eta = 0.2$, batch 64) and GD use gradients, and Adam ($\eta = 10^{-3}$) uses $u_k = \hat{m}_k/(\sqrt{\hat{v}_k} + \epsilon)$. Time is $t_K = \eta K$, $\|v\|_M^2 = v^\top M v$ and $\|v\|_{M^{-1}}^2 = v^\top M^{-1} v$.

B.1 Displacement from initialization as a function of time

Theorem B.1 (Theorem B1.1, Corollary B1.2, Proposition B1.3, Lemma B1.4)**.** *[Proved]*

(a) Gradient flow. *Let θ be absolutely continuous, with $\theta' = -Mv$ where $v(s)$ is a Clarke subgradient of L_S, and assume the chain rule $\frac{d}{ds} L_S(\theta) = \langle v, \theta' \rangle$ a.e. Then*

$$\|\theta(t) - \theta(0)\|_{M^{-1}} \le \int_0^t \|\theta'\|_{M^{-1}} \le \sqrt{t\,(L_S(\theta(0)) - L_S(\theta(t)))} \le \sqrt{tL_0}.$$

Per group, $\|\theta_g(t) - \theta_g(0)\|^2 \le m_g\, t\, E_g$, where $E_g = \int_0^t m_g \|v_g\|^2$ and $\sum_g E_g = L_S(\theta(0)) - L_S(\theta(t))$.

(b) Any update. *$\|\theta_K - \theta_0\|_{M^{-1}} \le \eta \sum_{k<K} \|u_k\|_M \le \sqrt{t_K E_K}$, where $E_K = \eta \sum_{k<K} \|u_k\|_M^2$.*

(c) GD under a descent condition. *If $L_S(\theta_{k+1}) \le L_S(\theta_k) - c\eta \|\nabla L_S(\theta_k)\|_M^2$ for all $k < K$, with $c > 0$ (condition (D)), then $E_K \le (L_0 - L_S(\theta_K))/c$.*

(d) SGD. *If L_S is β-smooth in the M-geometry on a convex set containing the iterates, $\eta \le 1/\beta$, and the minibatch gradients (with B indices drawn uniformly with replacement, independently of the past) have conditional M-variance at most ν^2/B, then $\mathbb{E}\|\theta_K - \theta_0\|_{M^{-1}}^2 \le 4t\,(L_0 - \mathbb{E}L_S(\theta_K)) + 2\eta t \nu^2 (1 + \beta t)/B$.*

(e) Adam. *If $\beta_1 \in [0, 1)$, $\beta_2 \in (0, 1)$ and $r = \beta_1^2/\beta_2 < 1$, every gradient sequence gives $|\hat{m}_t|/\sqrt{\hat{v}_t} \le C_t := \frac{1-\beta_1}{1-\beta_1^t}\big(\frac{1-\beta_2^t}{1-\beta_2} \cdot \frac{1-r^t}{1-r}\big)^{1/2}$, with equality for $g_s \propto (\beta_2/\beta_1)^s$. Hence $\|\Delta\theta_g\|_\infty \le \eta m_g S_T$ with $S_T = \sum_{t \le T} C_t$.*

Sketch. (a) $\|\theta'\|_{M^{-1}}^2 = v^\top M v = -\frac{d}{ds} L_S$; then the triangle inequality and Cauchy–Schwarz in s. (b)–(c) are the discrete analogues (sum (D)). (d) Split into drift and martingale noise; bound the drift by the descent lemma. (e) Cauchy–Schwarz on $m_t = (1 - \beta_1) \sum_s \beta_1^{t-s} g_s$ with weights $(\beta_1/\sqrt{\beta_2})^{t-s}$.

Weak or vacuous. [Numerical] For ReLU no global β exists, so (D) must be checked along the run: in a GD run at the study's rate (MAJ_3, $n = 2048$, $\eta = 0.2$) it holds only for 20 steps ($c = 0.26$), at $\eta = 0.02$ throughout ($c \ge 0.993$); the smoothness of (d) fails. In (e), $C_\infty = 7.27$ for $(\beta_1, \beta_2) = (0.9, 0.999)$, not the folklore 1 (real runs reach 1.5–1.7 in the first escape), and the a priori $\|\Delta W_1\|_F \le 8.9 \times 10^3$ after 20000 steps compares with a measured 4–11. For SGD at step 5000, $\|\Delta\|_{M^{-1}} = 3.7$ against $\Lambda = 85$ and $\sqrt{tE} = 159$. *Heuristic:* time-only bounds degrade as the initialization scale $\alpha \to 0$ although the learned function does not change.

B.2 Norm-based uniform convergence is vacuous here

For a complexity functional C with $\mathfrak{R}_n(\{f_\theta : C(\theta) \le \phi\}) \le \kappa\phi/\sqrt{n}$, the margin bound with union bounds over a norm grid $\phi_0 q^j$ and a margin grid γ_k (Theorem B2.1) has complexity term $c(\theta)/\gamma_k$, where $c(\theta) = 2\kappa\phi_0 q^{j(\theta)}/\sqrt{n}$ and $\phi_0 q^{j(\theta)} \ge C(\theta)$ is the grid round-up. With biases included, the derivation proves $\kappa_{\mathrm{path}} = \sqrt{2\ln(8(d+1))} = 3.32$ for the ℓ_1 path norm ϕ_1 (Proposition B2.2) and $\kappa_F = \sqrt{d+1}(1 + \sqrt{4\ln 2}) = 14.84$ for $P_F = \|a\|(\|[W_1\, b_1]\|_F \|W_2\|_F + \|b_2\|)$ (Proposition B2.3).

Lemma B.2 (Lemma B2.1a)**.** *[Proved] Let $\bar{m}_+(\theta) = \frac{1}{n}\sum_i (y_i f_\theta(x_i))_+$. If $c(\theta) \ge \max(1, \bar{m}_+(\theta))$, the margin bound of Theorem B2.1 is at least 1 at every margin, whatever the largest grid margin $\gamma_{\max}$.*

Sketch. The ramp loss satisfies $\varphi_\gamma(u) \ge 1 - u_+/\gamma$, so $\hat{R}_\gamma \ge 1 - \bar{m}_+/\gamma$; treat $\gamma \le 1$ and $\gamma > 1$ separately.

Numerical: vacuous everywhere. [Numerical] All 49 final Adam networks of the sweep ($n \in \{256, 2048, 16384\}$) have $\bar{m}_+ = 0.99$–1.01 and satisfy the lemma for the path-norm and Frobenius bounds; the spectral bound is vacuous too. The closest cases are $c_{\text{path}} = 1.46$ (x_1, $n = 16384$) and $c_F = 3.7$; the spectral complexity at unit margin is at least 679. The initialization's norms alone, $\phi_1(\theta_0) \approx 31.5$ and $P_F(\theta_0) \approx 11.3$, force $c \geq 1$ for $n \lesssim 4.4 \times 10^4$ and 1.1×10^5. *Heuristic reading:* the random features of the initialization dominate the norms (though training can lower them: 26 against 32 for x_1).

B.3 PAC-Bayes centred at the initialization

Theorem B.3 (Theorem B3.2)**.** *[Proved] Let Σ be a grid of G per-group noise levels $\sigma = (\sigma_g)$, and let $Q_{\theta,\sigma} = \mathcal{N}(\theta, \operatorname{diag}(\sigma_g^2 I_{p_g}))$ and $P_\sigma = \mathcal{N}(\theta_0, \operatorname{diag}(\sigma_g^2 I_{p_g}))$. Let $n \geq 8$. With probability $\geq 1 - \delta_0$, for all θ (however obtained) and all $\sigma \in \Sigma$,*

$$R(Q_{\theta,\sigma}) \leq \mathrm{kl}_+^{-1}\Big(\hat{R}_S(Q_{\theta,\sigma}),\ \tfrac{1}{n}\big[\mathrm{KL} + \ln \tfrac{2G\sqrt{n}}{\delta_0}\big]\Big), \qquad \mathrm{KL} = \sum_g \frac{\|\theta_g - \theta_{0,g}\|^2}{2\sigma_g^2}. \tag{PB}$$

Derandomized: *once θ, σ are fixed, let $\hat{e}$ be the mean error of m fresh draws $\theta' \sim Q_{\theta,\sigma}$ on batches from S, and $\hat{\Delta}$ their mean sign disagreement with f_θ on fresh unlabelled inputs. With probability $\geq 1 - \delta_0 - \delta_1 - \delta_2$, $R(\theta)$ is at most (PB) with $\hat{R}_S(Q)$ replaced by $\mathrm{kl}_+^{-1}(\hat{e}, \ln(1/\delta_1)/m)$, plus $\mathrm{kl}_+^{-1}(\hat{\Delta}, \ln(1/\delta_2)/m)$.*

Theorem B.4 (Time-based KL; Theorem B3.3)**.** *[Proved] With $\lambda_\sigma = \max_g m_g/2\sigma_g^2$, the KL in (PB) at the trained parameters satisfies: (a) for gradient flow ($c = 1$) or GD under (D), $\mathrm{KL} \leq t(L_0 - L_S(\theta(t)))\lambda_\sigma/c \leq tL_0\lambda_\sigma/c$, linear in t; (b) for any update, $\mathrm{KL} \leq t_K \sum_g m_g E_{K,g}/2\sigma_g^2$ with the measured per-group energies; (c) for Adam, $\mathrm{KL} \leq \sum_g p_g(\eta m_g S_K)^2/2\sigma_g^2$, quadratic in K since $S_K \geq K$; (d) for SGD under the hypotheses of Theorem B.1(d), with $V(t) = 4tL_0 + 2\eta t\nu^2(1+\beta t)/B$: with probability $\geq 1 - \delta_0 - \delta'$, (PB) holds at θ_K with $\mathrm{KL} \leq \lambda_\sigma V(t)/\delta'$; and for the mixture "rerun SGD, then perturb" the PAC-Bayes-kl inequality holds with $\mathrm{KL} \leq \lambda_\sigma V(t)$.*

Sketch. The prior depends only on θ_0, which is independent of S, so the PAC-Bayes-kl bound (Maurer 2004; in the line of [23]) applies, with a union bound over Σ. Fresh Monte Carlo draws (as in [16]) and the pointwise inequality $\mathbb{1}[yf_\theta \leq 0] \leq \mathbb{1}[yf_{\theta'} \leq 0] + \mathbb{1}[\operatorname{sgn} f_{\theta'} \neq \operatorname{sgn} f_\theta]$ derandomize it. For Theorem B.4, $\mathrm{KL} \leq \lambda_\sigma \|\theta - \theta_0\|_{M^{-1}}^2$; insert Theorem B.1, and for (d) Markov's inequality or convexity of KL.

Numerical. [Numerical] On the final Adam networks the deterministic bound is 0.26–0.56 at $n = 16384$ for the five targets with ≤ 3 relevant variables, 0.55–1.0 at $n = 2048$, vacuous at $n = 256$, and uninformative (0.79–1.0) with 4–6 relevant variables even at $n = 16384$. The KL for x_1 at $n = 16384$ is 1804 nats, against about 4 nats to describe x_1 as a circuit. After 100–300 SGD or Adam steps the bound is 0.10–0.34 (best 0.11 for x_1, 0.23 for MAJ_3, with a union bound over 10 checkpoints). The time-based forms are informative (< 0.5) only up to $t \approx 4$ or 50 Adam steps: (a) gives 0.59–0.77 for GD at $\eta = 0.02$; (b) gives 0.39 for SGD on x_1 at step 20 and ≥ 0.9 from step 100, and is always weaker than (PB) with the measured displacement; (c) gives 0.44–0.56 at step 50 and is vacuous by step 200; (d) is not evaluated, since smoothness fails for ReLU.

B.4 Distillation certificates without labelled hold-out data

Setting. $\{\mathcal{H}_c\}_{c \in I}$ is a countable family of finite classes $\{\pm 1\}^d \to \{\pm 1\}$, with weights $\sum_c \pi(c) \leq 1$, all fixed in advance. The predictors to certify, $h_1, \ldots, h_J$, are arbitrary functions of S and of training

randomness $\xi \perp S$. The unlabelled sample U of m uniform inputs is independent of (S, ξ).

Theorem B.5 (Theorem C1). *[Proved] With probability $\geq 1-\delta_1-\delta_2$, simultaneously for all $j \leq J$, all $c \in I$ and all $g \in \mathcal{H}_c$ (so also for g selected from S, ξ, h_j and U),*

$$\operatorname{err}(h_j) \leq \operatorname{kl}_+^{-1}\Big(\widehat{\operatorname{err}}_S(g), \frac{\ln|\mathcal{H}_c| + \ln\frac{1}{\pi(c)} + \ln\frac{1}{\delta_1}}{n}\Big) + \operatorname{kl}_+^{-1}\Big(\widehat{\operatorname{dis}}_U(h_j, g), \frac{\ln|\mathcal{H}_c| + \ln\frac{1}{\pi(c)} + \ln\frac{J}{\delta_2}}{m}\Big).$$

Variants: *with K candidates chosen using part U_A of the pool and disagreement measured on an independent part U_B (m_B points), the second complexity term may be $\ln(JK/\delta_2)/m_B$; an exactly computed disagreement may be used as is.*

Sketch. The Occam event, a kl-Chernoff and union bound over $\bigcup_c \mathcal{H}_c$ [7], does not mention h or the selection. Conditionally on (S, ξ), U is still iid. Then $\operatorname{err}(h) \leq \operatorname{err}(g) + \operatorname{dis}(h, g)$. The bound breaks if the disagreement is measured on training inputs, or if the network is trained on U.

The classes, counted. [Proved] Let $\chi_S = \prod_{i\in S} x_i$, let D_k^* count the nonempty characters of degree $\leq k$, and let $\Phi(N, T) = 2\sum_{i\leq T}\binom{N-1}{i}$ (Cover's count). The signs $\mathcal{F}_{k,T}$ of $w_0 + \sum_{j\leq T} w_j \chi_{S_j}$, with distinct S_j of size 1–k, satisfy $\ln|\mathcal{F}_{k,T}| \leq \ln\binom{D_k^*}{T} + \ln\Phi(2^{\min(T,d)}, T)$ (Proposition C1.2). Fan-in-2 circuits with at most T gates compute at most $\sum_{s\leq T}(16(d+s)^2)^s(d+s)/s!$ functions (Proposition C1.3). Juntas are counted exactly: $|\mathcal{J}_{\leq k}| = \sum_{j\leq k}\binom{d}{j}N_j$ with $N_j = \sum_{i\leq j}(-1)^{j-i}\binom{j}{i}2^{2^i}$ (Proposition C1.4). At $d = 30$, $\ln|\mathcal{J}_{\leq 3}| = 13.7$ and $\ln|\mathcal{C}_{\leq 8}| = 73.4$ nats.

Numerical. [Numerical] With juntas and Fourier signs (circuits were not searched) and selection from unlabelled inputs only, every target's saved final Adam network is certified below 0.5 at $n = 2048$ for every seed, 9 of 13 targets below 0.04, and every network at $n = 16384$ at ≤ 0.021. Fourier signs certify MAJ_9 at 0.034 and a 10-variable threshold at 0.09–0.10 ($n = 2048$), where the best junta certificates are 0.41–0.43 and 0.20. No certificate falls below the true error. Along training, the class that minimizes the certificate grows in stages. *Weakness:* the certificate is a posteriori and does not say why training yields a near-circuit.

B.5 One-pass training: online-to-batch conversions

In the online protocol each step draws a fresh batch, the current predictor predicts on it, and then the network updates. Write ℓ_j for the mistake on sample j of the N samples, p_j for the test error of the predictor that saw it, and $M_N = \sum_j \ell_j$.

Proposition B.6 (Propositions B3.4, C4.1, C4.2). *[Proved] (1) With probability $\geq 1-\delta$, $\frac{1}{N}\sum_j p_j \leq \frac{M_N}{N} + \sqrt{\ln(1/\delta)/2N}$, and likewise for any window of steps fixed in advance (B3.4). For a finite grid Λ, with probability $\geq 1-\delta$, for all $\lambda \in \Lambda$,*

$$\tfrac{1}{N}\textstyle\sum_j p_j \leq (\lambda M_N + \ln(|\Lambda|/\delta))/((1-e^{-\lambda})N).$$

(2) For a finite set I of steps, let $\widehat{\operatorname{err}}_{\geq i}(h_i)$ be the error of h_i on the $N_{\geq i}$ samples of steps $\geq i$. With probability $\geq 1-\delta$, for all $i \in I$,

$$\operatorname{err}(h_i) \leq \operatorname{kl}_+^{-1}(\widehat{\operatorname{err}}_{\geq i}(h_i), \ln(|I|/\delta)/N_{\geq i}).$$

Adding the term $\operatorname{kl}_+^{-1}(\widehat{\operatorname{dis}}_U(h_{\text{final}}, h_i), \ln(|I|/\delta')/m)$, for m unlabelled inputs independent of the run, and minimizing over i bounds $\operatorname{err}(h_{\text{final}})$ with probability $\geq 1-\delta-\delta'$. Without such a term a small average error says nothing about the final iterate.

Sketch. $p_j - \ell_j$ are martingale differences, so Hoeffding's lemma applies; the fast rate uses the supermartingale $\exp\sum_j[(1-e^{-\lambda})p_j - \lambda\ell_j]$ [10]. For (2), later samples are fresh for h_i.

Numerical. [Numerical] On one-pass Adam runs with logged progressive errors, B3.4 is within 0.006 of the measured window-average test error, and C4.1 certifies the second half of training at ≤ 0.0017 in all 18 reruns (`study/ONLINE-BOUNDS.md`). Part (2) and the fast-rate form are not evaluated.

B.6 Two-layer ReLU networks from balanced small initialization

Setting (Part I of D). $f(x) = \sum_{j \leq m} a_j \sigma(w_j \cdot \tilde{x})$ with $\tilde{x} = (x, 1)$, $L = \frac{1}{2}\|y - f\|_n^2$, $r = y - f$, and λ_X the top eigenvalue of $\frac{1}{n}\sum_i \tilde{x}_i \tilde{x}_i^\top$. The utility of a unit direction is $U_r(v) = \langle r, \sigma(v \cdot \tilde{x})\rangle_n$, $U^*(r) = \sup_v |U_r(v)|$ and $U_0^* = U^*(y)$. The initialization is balanced, $|a_j(0)| = \|w_j(0)\| = \alpha$ with signs ϵ_j; the masses are $\rho_j = a_j^2 + \|w_j\|^2$ and $P = \sum_j \rho_j$, so the width enters only through $P(0) = 2m\alpha^2$. A *selection flow* is gradient flow with any measurable choice of subgradients at the kinks (every Clarke flow is one). Unless stated otherwise, the statements below concern selection flows from this initialization, and hold for any data and width.

Lemma B.7 (Lemma D1(b), (c), (g))**.** *[Proved] (i)* Mass clock: *with* $\hat{w}_j = w_j/\|w_j\|$, $\rho_j(t) = \rho_j(0)\exp\big(2\int_0^t \epsilon_j U_{r(s)}(\hat{w}_j(s))\,ds\big)$. *(ii)* Measure form: $f_t = \sum_j \epsilon_j \frac{\rho_j}{2}\sigma(\hat{w}_j \cdot \tilde{x})$ *everywhere, so the variation norm is at most* $P/2$. *(iii)* Merging law: *for* $\epsilon_j = \epsilon_k$ *let* $Q_{jk} = a_j a_k - w_j \cdot w_k = \|w_j\|\|w_k\|(1 - \cos\angle(w_j, w_k))$. *(g1)* Q_{jk} *is constant while* j *and* k *use the same subgradient selections at every training point. (g2) It is nonincreasing while* $\epsilon_j r_i \leq 0$ *at every training point separating them. (g3) In general* $\frac{d}{dt}\sqrt{Q_{jk}} \leq \sqrt{(d+1)\|w_j\|\|w_k\|/2}\,\bar{r}_{jk}^{\mathrm{rep}}$, *where* $\bar{r}_{jk}^{\mathrm{rep}}$ *is the residual mass* $\frac{1}{n}\sum|r_i|$ *over separating points with* $\epsilon_j r_i > 0$, *wherever* $Q_{jk} > 0$.

Sketch and credit. $a_j^2 - \|w_j\|^2$ is conserved since $s\,z = \sigma(z)$ for every subgradient s, and positive homogeneity gives $\dot{\rho}_j = 2\epsilon_j \rho_j U_r(\hat{w}_j)$. By monotonicity of $\partial\sigma$, $\dot{Q}_{jk}$ is a sum over separating points of $\epsilon_j r_i$ times a nonnegative factor. (g1) is the off-diagonal entry of the classical balancedness law $W_1 W_1^\top - W_2^\top W_2 = \text{const}$ of two-layer linear networks [4, 15], applied to a pair that shares its selections and is therefore locally a linear network. Only (g2) and (g3) may be new.

Theorem B.8 (No growth before $\ln(1/\alpha)/U_0^*$; Theorem D2)**.** *[Proved] (1) The path norm satisfies* $P(t) \leq P(0)\exp(2\int_0^t U^*(r(s))ds)$, *and* $\sup_{x\in\{\pm1\}^d}|f_t(x)| \leq \frac{\sqrt{d+1}}{2}P(t)$. *(3) For* $\beta > P(0)$ *the escape time* $t_\beta = \inf\{t : P(t) \geq \beta\}$ *satisfies*

$$t_\beta \geq T_\beta := \frac{\ln(\beta/P(0))}{2(U_0^* + \lambda_X \beta/2)},$$

so for $t \leq T_\beta$ *the network is within* $\frac{\sqrt{d+1}}{2}\beta$ *of the zero function on the whole cube. (6) Every selection flow has* $\liminf_{\alpha\to 0} t_\beta U_0^*/\ln(1/\alpha) \geq 1$, *at every fixed* $\beta > 0$ *and width, and the limit is* 1 *(for every* $\beta < \sqrt{2}$*) for* $y = x_1$ *on the full cube with one neuron started at the maximizer of* U_y *and* $\sigma'(0) = 0$. *So the leading constant* $1/U_0^*$ *is sharp.*

Sketch. Use (i) with $|U_r| \leq U^*(r)$ and $U^*(r_t) \leq U_0^* + \sqrt{\lambda_X}\|f_t\|_n < U_0^* + \lambda_X\beta/2$ for $t < t_\beta$. In the instance $P(t)$ is logistic; the $\liminf$ follows from $t_\beta \geq t_{\beta'} \geq T_{\beta'}$ with $\beta' \to 0$. Lower-order terms are not claimed sharp; that the escape actually happens near T_β is not proved. The derivation also gives a discrete-GD form and a certified bound $\hat{U}(r) \geq U^*(r)$ from two eigenvalue computations (Lemma D2.1).

Theorem B.9 (Theorems D3 and D3.2)**.** *[Proved]* Path norm. *For every* $\beta > P(0)$, $P(t) \leq \beta + \|y\|_n^2(t - T_\beta)_+$ *(a priori) and* $\sqrt{P(t)} \leq \sqrt{\beta} + \int_{T_\beta}^t \sqrt{-\dot{L}}$ *(from the training-loss curve).* Generalization.

Let the inputs be uniform, $b^2 = \frac{1}{2}(1+\lambda_X)$ and $e_K = 2B^2(b+1)^2/(K\gamma^2)$. Fix, before seeing data, a margin grid Γ and, for each K, a net $\mathcal{N}_K$ of the unit sphere of $\mathbb{R}^{d+1}$ of resolution $1/\sqrt{K(d+1)}$, with $|\mathcal{N}_K| \le (1+2\sqrt{K(d+1)})^{d+1}$. With probability $\ge 1-\delta$, simultaneously for every two-layer ReLU network f of any width (balanced or not), every $K \ge 1$, $\gamma \in \Gamma$ and dyadic $B = 2^\ell \ge \sum_j |a_j|\|w_j\|$,

$$\operatorname{err}(\operatorname{sgn} f) \le \mathrm{kl}_+^{-1}\Big(\widehat{P}_S[yf \le 2\gamma] + e_K,\ \frac{K\ln(2|\mathcal{N}_K|+1) + \ln\frac{2|\Gamma|K(K+1)(|\ell|+1)(|\ell|+2)}{\delta}}{n}\Big) + e_K.$$

Sketch. For the path norm, $\dot{P} = 4\langle r, f\rangle_n \le \|y\|_n^2$ and $\frac{d}{dt}\|\theta\| \le \sqrt{-\dot{L}}$. For generalization, Maurey sampling of K neurons with weights $|a_j|\|w_j\|$ (the measure of Lemma B.7(ii) under balance), rounding to a net and Chebyshev's inequality under the test and training distributions give K-term approximants at margin cost e_K; then apply kl-Occam. *Vacuous:* at $d = 16$ and $B \approx 1.4$ the bound needs $n \gtrsim 10^5$, and it is 1.0 in all runs below.

The non-vacuous version is a posteriori: Proposition D3.3 ([Proved]) is a kl-Occam bound at margin γ, uniform over K, B, γ, for every predictor within $\gamma/4$ in sup norm of some $\sum_{k\le K} c_k\sigma(v_k\cdot\tilde{x})$ with $\sum_k |c_k| \le B$, at capacity $K[(d+1)\ln(1+16B\sqrt{d+1}/\gamma) + \ln(2\lceil 8KB\sqrt{d+1}/\gamma\rceil + 1)]$ (about 100 nats per neuron at $d = 16$, $B \approx 1.4$, $\gamma = \frac{1}{2}$).

Proposition B.10 (Persistence of clusters; Proposition D4.1)**.** *[Conditional] Fix t_1, same-sign clusters G_k with reference neurons j_k, and a remainder D with mass P_D. Suppose that on $[t_1, t]$:*

(A) every training point separating some $j \in G_k$ from j_k has $\epsilon_j r_i \le 0$, for almost every s.

Then $g_t = \sum_k c_k\sigma(\hat{w}_{j_k}\cdot\tilde{x})$, with $c_k = \sum_{G_k}\epsilon_j\rho_j/2$ and $\sum|c_k| \le P(t)/2$, satisfies

$$\sup_x |f_t - g_t| \le \sqrt{d+1}\Big[\sqrt{2}\sum_k\sum_{j\in G_k}\frac{\|w_j(t)\|^{3/2}}{\|w_{j_k}(t)\|^{1/2}}\sqrt{Q_{jj_k}(t_1)} + \tfrac{1}{2}P_D(t_1)e^{2\int_{t_1}^t \hat{U}(r_s)ds}\Big].$$

Without (A), add $\int_{t_1}^t \sqrt{(d+1)\|w_j\|\|w_{j_k}\|/2}\,\bar{r}^{\mathrm{rep}}_{jj_k}$ to each $\sqrt{Q_{jj_k}(t_1)}$. When the right-hand side is at most $\gamma/4$, Proposition D3.3 applies at time t with this K.

Sketch. Under (A), (g2) keeps $Q_{jj_k} \le Q_{jj_k}(t_1)$. The dormant neurons obey the clock bound from t_1 on. (A) can be checked from the training residuals, but it is not proved.

Conjecture B.11 (Conjecture D4)**.** *[Conjecture] Fix S and the width, take initial directions in general position, and let $\alpha \to 0$. Let K be the number of selection-critical directions of $\pm U_y$ that attract a positive fraction of initial directions. Suppose the K-neuron least-squares fit on them has residual $r^\star$ with $\epsilon r_i^\star < 0$ at the training points on the clusters' boundary hyperplanes. Then $\sup_{t\le(1+c)\ln(1/\alpha)/U_0^*}\inf_g\sup_x|f_t - g| \to 0$ over K-neuron functions g, for every $c < c^\star(S)$, where $c^\star\ln(1/\alpha)/U_0^*$ is the time to the next escape.*

Numerical. [Numerical] ($d = 16$, $n = 1000$–4000, width 64, $\alpha = 10^{-6}$–10^{-2}, small-step GD, exact test error on all 2^{16} inputs.) The growth bound reaches 0.84–0.95 of its exponent, and at $\alpha \le 10^{-3}$, T_β is 60–81% of the observed escape time. At $\alpha \le 10^{-3}$ the neurons condense into $K = 2$ clusters ($1 - \cos \le 2\cdot 10^{-4}$; one neuron stays out at $n = 4000$) that, except for x_1, whose residual vanishes, sit on data hyperplanes (training margins 10^{-12}–$3\cdot 10^{-7}$); where (A) holds, Q grew by at most 0.2%. D3.3 with the network's two clusters certifies MAJ_3 at 0.19 ($n = 1000$) and 0.051 ($n = 4000$), true error 0 (computed for the balanced surrogate of the GD network). On $x_1 + \frac{1}{2}x_2x_3$ at $\alpha = 10^{-6}$ every neuron is recruited by the first escape and the network stalls, the "two-edged sword" of [8].

B.7 Quadratic networks with fixed output signs

Setting (Part II of D). $f(x) = \langle X, xx^\top - I\rangle$ with $X = UU^\top - VV^\top$: centred quadratic units with output weights fixed at ± 1, i.e. symmetric matrix sensing (trainable output weights are not covered). The flow is $\dot U = 2M_rU$, $\dot V = -2M_rV$ with $M_r = \frac{1}{n}\sum_i r_i(x_ix_i^\top - I)$, from $U(0) = \alpha U_0$, $V(0) = \alpha V_0$, $s_0 = \max(\|U_0\|_{op}, \|V_0\|_{op})$. $N_\epsilon(t)$ counts the eigenvalues $\geq \epsilon$ of $UU^\top$ and of $VV^\top$; $\Theta_\epsilon = \frac{1}{4}\ln(\epsilon/\alpha^2 s_0^2)$, $\tau(t) = \int_0^t \|r\|_n$ and $P = \|U\|_F^2 + \|V\|_F^2$. The coherence $\kappa = 2\lambda_{\max}(\frac{1}{n}\Phi^\top\Phi)$, with $\Phi_{i,(k,l)} = x_{ik}x_{il}$ for $k < l$, depends only on the training inputs (not on labels, trajectory or width); its population value is 2.

Theorem B.12 (Theorem D6, with Theorem D5)**.** *[Proved] For gradient flow, all t and every ϵ with $\Theta_\epsilon > 0$: (2) $N_\epsilon(t) \leq \kappa\,(\tau(t)/\Theta_\epsilon)^2$; (3) the first time t_ϵ at which an eigenvalue reaches ϵ satisfies $t_\epsilon \geq \Theta_\epsilon/(\|M_y\|_{op} + 2\kappa\sqrt{d}\,\epsilon)$; (4) $P \geq \|X\|_*$ and $\dot P = 4\langle r, f\rangle_n \leq \|y\|_n^2$. (5) For GD and SGD with step matrices M_j (full-batch or minibatch B_j): if $2\eta\|M_j\|_{op} < 1$ at every step, (2)–(3) hold for GD with $\tau_J = \eta\sum_{j<J}\|r_j\|_n$; for SGD, (2) holds as $N \leq (\eta\sum_j \sqrt{\kappa_{B_j}}\|r_j\|_{B_j})^2/\Theta_\epsilon^2$ with κ_{B_j} the batch coherence, and (3) with the maxima of $\|M_y^{B_j}\|_{op}$ and κ_{B_j} over the batches before first growth. With no step-size condition, $P_J \leq e^{4\eta^2\sum_j\|M_j\|_{op}^2}(P(0) + \eta\sum_j\|y\|_{B_j}^2)$.*

The minibatch (SGD) form of (3) was added after the second referee pass and has not been refereed. *Sketch.* The log-det clock (Theorem D5; Ky Fan's maximum principle for the flow, Horn's inequality for GD and SGD) gives $\sum_{k\leq K}\ln\sigma_k(U(t)) \leq \sum_{k\leq K}\ln\sigma_k(U(0)) + 2\int_0^t\sum_{k\leq K}\lambda_k(M_{r(s)})ds$, hence $N\Theta_\epsilon \leq \int(\text{two Ky Fan sums of } \pm M_r)$. Let $Z = \Pi_+ - \Pi_-$ project onto the relevant eigenvectors of M_r, so that $\|Z\|_F^2 \leq N$. Then each instant contributes at most $\langle M_r, Z\rangle = \langle r, \mathcal{A}(Z)\rangle_n \leq \|r\|_n\sqrt{\kappa N}$. So $N\Theta_\epsilon \leq \tau\sqrt{\kappa N}$. *Weight-free, not a priori:* τ is read off the loss curve. The time-only form $\tau \leq \|r(0)\|_n t$ (flow) is much weaker; for x_1x_2 below it gives $N \leq 42$ at $t = 20$, against 4.

Theorem B.13 (Theorem D7)**.** *[Proved] Let $d \geq 2$, $y \in \{\pm 1\}$, and fix a margin grid $\Gamma \subset (0, 8(d-1)]$. For $K \geq 0$, dyadic $B = 2^\ell$ and $\gamma \in \Gamma$, put $\eta_\lambda = \gamma/(8(d-1)\max(K,1))$ and $\eta_z = \gamma/(16(d-1)(B+1))$, and let $\mathcal{G}_{K,B,\gamma}$ be the matrices $\sum_{k\leq K}\lambda'_k z_k z_k^\top$ with $\lambda'_k \in \eta_\lambda\mathbb{Z}$, $|\lambda'_k| \leq \eta_\lambda\lceil B/\eta_\lambda\rceil$ and z_k in a fixed η_z-net of S^{d-1}, so that $C_{K,B,\gamma} := \ln|\mathcal{G}_{K,B,\gamma}| \leq K\ln(2\lceil B/\eta_\lambda\rceil + 1) + dK\ln(1 + 2/\eta_z)$. Put $\epsilon_\gamma = \gamma/(8(d-1))$. With probability $\geq 1-\delta$, simultaneously for all t, all $\gamma \in \Gamma$, every integer $N \geq N_{\epsilon_\gamma}(t)$ and every dyadic $B \geq P(t)$,*

$$\operatorname{err}(\operatorname{sgn} f_t) \leq \mathrm{kl}_+^{-1}\Big(\widehat{P}_S[yf_t \leq \gamma],\ \frac{C_{N,B,\gamma} + \ln\frac{2|\Gamma|(N+1)(N+2)(|\ell|+1)(|\ell|+2)}{\delta}}{n}\Big).$$

The event is the same for every (U, V), so it covers all GD and SGD iterates. Admissible choices include the measured $N_{\epsilon_\gamma}(t)$ and $P(t)$, $N = \lfloor\kappa(\tau/\Theta_{\epsilon_\gamma})^2\rfloor$ for the flow (for $\Theta_{\epsilon_\gamma} > 0$; for GD and SGD, the forms of Theorem B.12(5)), and B from Theorem B.12(4) (flow) or (5) (GD and SGD).

Sketch. Split X into the eigencomponents $\geq \epsilon_\gamma$ of $UU^\top$ and $VV^\top$ (rank $\leq N$, nuclear norm $\leq B$) and the rest. Since $\|xx^\top - I\|_* = 2(d-1)$, the rest moves f by at most $\gamma/4$. Rounding into $\mathcal{G}$ costs less than $\gamma/4$; then apply kl-Occam at margin $\gamma/2$. The capacity is 127–129 nats per unit at $d = 16$, $B = 1$, $\gamma = \frac{1}{2}$, so the bound beats the unrestricted degree-2 class (about 964 nats) only for $N \leq 7$.

Numerical. [Numerical] ($d = 16$, width $16 + 16$, $\alpha = 10^{-6}$, GD with $\eta = 0.01$ and one SGD run with batch 64, exact test error.) The clock holds to 10^{-8} and the rank bound is within a factor 1.7–2.7 of the truth. For x_1x_2 at $n = 3000$, D7 gives 0.084 with the measured rank and 0.213 with the weight-free rank and the proved GD norm bound, against 0.276 for the unrestricted class (true

error 0); the step-size condition holds ($2\eta\|M_j\|_{op} \le 0.021$). On MAJ of three pairs the residual plateaus above zero, so τ keeps growing: the weight-free rank bound rises from 23 to 47 while the true rank stays 6, and D7 becomes vacuous (0.969). For the SGD run on the same target, the proved SGD forms give D7 bounds of 0.89–0.90, with N capped at the trivial 32.

C Experimental details

Network training (`study/nn_train.py`).

- **Architecture.** $f(x) = a^\top \mathrm{relu}(W_2 \mathrm{relu}(W_1 x + b_1) + b_2)$, widths $N \times N$.
- **Initialization.** $W_1 \sim \mathcal{N}(0, 1/d)$, $W_2 \sim \mathcal{N}(0, 1/N)$, $a \sim \mathcal{N}(0, 1/N^2)$, zero biases.
- **Adam** [19] with $\beta = (0.9, 0.999)$, $\epsilon = 10^{-8}$ and learning rate 10^{-3}. The per-layer multipliers, relative to base width 64, are 1 for W_1, b_1 and $64/N$ for W_2, b_2, a.
- **SGD:** learning rate 0.2, with multipliers $N/64$ for W_1, b_1, 1 for W_2, b_2 and $64/N$ for a.
- **Loss and data.** Square loss $\frac{1}{2}(f-y)^2$; minibatch 64, sampled with replacement from a fixed training set.
- **Runs.** Training sets are nested across n for each seed, and all (n, seed) networks are trained in parallel. The checkpoints are at steps 0, 10, 20, 50, 100, 200, 500, 1000, 2000, 3000, 5000, 7000, 10000, 15000 and 20000. The test set has 4096 fresh points.
- **Recorded quantities:** the network's Fourier coefficients on all characters of the relevant variables up to degree 4; the participation ratios of both layers' kernels; weight norms and displacements; and the test predictions.

Baselines.

- **NNGP kernel:** the infinite-width kernel of the same architecture at initialization, evaluated in closed form on the cube, used in kernel ridge regression with ridge $10^{-4}\,\mathrm{tr}K/n$, for $n \le 4096$.
- **Occam over juntas:** the smallest $k \le 4$ such that some k-subset of coordinates is consistent with the training data, predicting by its truth table, with unseen cells given the majority label. If no subset is consistent, the 4-subset with the fewest training mistakes is used.
- **sswm:** strong-selection/weak-mutation evolution of a single shared circuit of fan-in-2 gates (BUF, signed AND, XOR) with a ridge readout. One random local mutation is proposed per tick: adding one gate (weight 1), a chain of two or three fresh gates (weights ϵ, ϵ^2), or rewiring (1/4), retyping (1/8) or deleting (1/4) a leaf gate. A mutation that adds gates is kept if it lowers the training objective by more than a significance threshold per fresh gate; rewires and retypes are kept if they lower it at all, and deletes never are.

The race model (`study/models/race.py`). The settings are $\theta = 1.5$, ridge $\lambda = 10^{-3}$, integer weights in $\{-3, \dots, 3\}$, ReLU gates, proto-unit orders up to $K = 4$ with $P = 8$ units per order, 8 power-iteration steps per firing, and at most 48 gates. Full details and the 30 ablations are in `study/ABLATIONS.md`.

The minimal model and predictor (`study/models/minimal.py`, `study/minimal_predict.py`).

- **Gates.** Depth-1 gates have fan-in at most 4; depth-2 gates have fan-in at most 3, over active literals and earlier gates.
- **Dense units.** A linear Hebbian unit, and one order-2 spectral unit per stage.
- **Acceptance.** The acceptance rule is (1), with $c = 2.25$ and $a = 18$.

The hybrid model (`study/models/hybrid.py`). The race model plus an NNGP background in a joint ridge readout, with background ridge $c_{\mathrm{bg}} n/t$, $c_{\mathrm{bg}} = 0.003$, and growth stopped when the training MSE falls below 0.03. Above $n = 2048$ the background is a rank-512 random-feature approximation; this is used in the scoreboard but not in the diagnosis.

Compute. Everything ran on a 4-core CPU container. A model evaluation over the full (21 targets) $\times$ (11 n) $\times$ (3 seeds) grid takes about 1.5–12.5 minutes single-threaded.

D How this paper was produced

This paper was written by an AI system, Claude (Anthropic), over several sessions on 23–24 October 2026. Kaarel Hänni prompted and steered it. No human has reviewed the proofs or the code.

Verification process.

- **Derivations.** Each of the four derivations behind Section 6, files `theory/A`–`D`, was read by an independent adversarial AI referee instructed to find errors. Each was revised, and then refereed a second time.
  - *Round 1.* Derivation A failed (one mislabelled claim, five major issues). B failed on two overclaims. C passed with three required revisions. D passed with minor issues.
  - *Round 2.* All four passed with only minor issues (7, 8, 10 and 7). These were then applied; for A, B and C, each set of fixes was checked by a further agent.
  - *Unrefereed addition.* The minibatch form of one item of D's Theorem D6(5) was added during that last step and has not itself been refereed.
  - *Records.* The referee reports are in `theory/checks/`.
- **Numbers.** The numbers in the summary write-up were checked against the generated data by independent fact-checking agents, and each flagged discrepancy was re-checked by a second agent before correction. Of 58 flags, 49 were confirmed and corrected.
- **Model write-ups.** The ablation, minimal-model and hybrid write-ups each had a separate verification pass that reproduced their numbers from the saved evaluations and corrected several mechanism claims.

Pre-registration. Two tests were run with predictions recorded before the network results were inspected: the $d = 120$ test of Section 4.3, and the composition test of Section 5. For the composition test the git history shows the order. For the $d = 120$ test it rests on the save times written into the prediction files (`study/understanding/`) and on the modification times of the network results, which are kept in the session scratch directory, not in the repository.

What remains unchecked.

- Literature claims rest on abstracts and memory of the papers, not on rereading them.
- The $d = 120$ section of the synthesis note has not had an independent verification pass beyond our recomputation of the networks' n^* and the timestamp check.

Reproducibility. All code is in `study/`: the network harness `nn_train.py` (JAX, CPU), the models in `models/`, and analysis scripts that write the generated reports (`*.md`). The logged run times of the network training sum to roughly six hours on the 4-core CPU. Raw network outputs were kept in a session scratch directory and are not in the repository; the scripts regenerate them.